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Explainable deep learning approach for advanced persistent threats (APTs) detection in cybersecurity: a review

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Abstract

In recent years, Advanced Persistent Threat (APT) attacks on network systems have increased through sophisticated fraud tactics. Traditional Intrusion Detection Systems (IDSs) suffer from low detection accuracy, high false-positive rates, and difficulty identifying unknown attacks such as remote-to-local (R2L) and user-to-root (U2R) attacks. This paper addresses these challenges by providing a foundational discussion of APTs and the limitations of existing detection methods. It then pivots to explore the novel integration of deep learning techniques and Explainable Artificial Intelligence (XAI) to improve APT detection. This paper aims to fill the gaps in the current research by providing a thorough analysis of how XAI methods, such as Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), can make black-box models more transparent and interpretable. The objective is to demonstrate the necessity of explainability in APT detection and propose solutions that enhance the trustworthiness and effectiveness of these models. It offers a critical analysis of existing approaches, highlights their strengths and limitations, and identifies open issues that require further research. This paper also suggests future research directions to combat evolving threats, paving the way for more effective and reliable cybersecurity solutions. Overall, this paper emphasizes the importance of explainability in enhancing the performance and trustworthiness of cybersecurity systems.

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1 Introduction

Advanced persistent threats (APTs) represent a significant cybersecurity challenge in the digital era. (Hasan et al. 2023). In this study, we explore the integration of Explainable Artificial Intelligence (XAI) with deep learning models to enhance the detection of Advanced Persistent Threats (APTs). APTs are prolonged and targeted cyberattacks in which an intruder gains access to a network and remains undetected for an extended period. Unlike traditional attacks, APTs aim to steal sensitive data over time rather than causing immediate damage. APT attacks are conducted by highly skilled adversaries with vast resources and target organizations through IT network systems for long-term access, without being discovered. APT attacks target critical infrastructure, such as financial institutions, government agencies, energy companies, shipping and transportation companies, industrial companies, and food services, which can disrupt critical infrastructure and cause massive damage. APT attacks have major impacts on countries, organizations, and individuals, leading to severe financial losses, reputational damage, legal liability, lost productivity, business continuity issues, and disruption of critical operations (Salim et al. 2023). For example , the Stuxnet worm, attributed to state-sponsored actors, targeted Iran's nuclear facilities and caused significant operational disruptions (Ahmad et al. 2024). Another notable example is the SolarWinds attack in 2020, which involved compromising the Orion software platform (Alkhadra et al. 2021). This attack affected numerous government agencies and private sector companies, leading to widespread data breaches and significant financial losses. The estimated cost of the SolarWinds breach is still being assessed, but the vulnerability of even the most secure networks to sophisticated APTs has been highlighted.

The National Institute of Standards and Technology (NIST) defines APTs as sophisticated, resourceful adversaries that use multiple attack methods, such as cyber-attacks, physical intrusions, and deception to infiltrate an organization's infrastructure (de Abreu et al. 2020). APTs are particularly challenging to detect due to their stealth and persistence, blending with legitimate network traffic and activities (Salim et al. 2023). Unlike typical malware, which executes its payload quickly, APTs focus on long-term espionage or data theft, making detection more difficult (Jabar and Mahinderjit Singh 2022).

In August 2018, Symantec[1] reported APT attacks by the Leafminer group, known as RASPITE, in the Middle Eastern region since 2017[2]. Figure 1 illustrates the distribution of APT attacks by the Leafminer group, across various industrial sectors. Government agencies were one of the primary targets, representing 17% of all attacks. Financial institutions were equally targeted, accounting for 17% of all attacks. The Ponemon Institute, as analyzed by IBM Security, reported that the average cost of an APT attack and data breach in 2023 was $4.35 million[3]. Large organizations experienced an average loss of $6.93 million per incident. This distribution emphasizes the need for robust cybersecurity measures

[1] https://symantec-enterprise-blogs.security.com/blogs/threat-intelligence/leafminer-espionage-middle-east.

[2] https://attack.mitre.org/groups/G0099/.

[3] https://www.ibm.com/reports/data-breach.

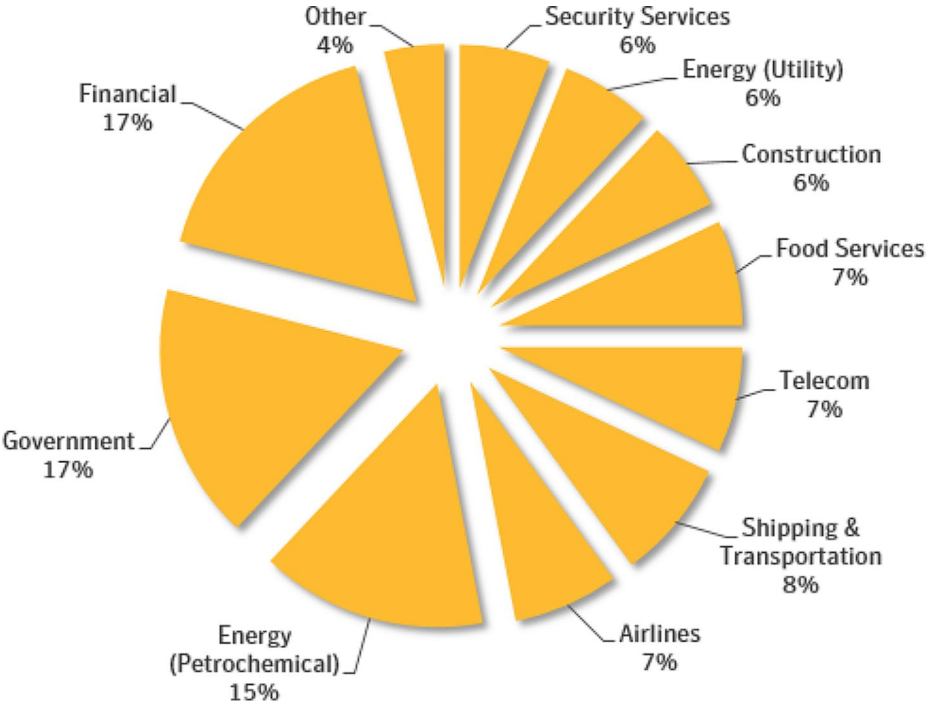


Fig. 1 APT attack distribution by Leafminer group. (Source: Symantec)

across diverse industries to protect against sophisticated threats, such as those posed by the Leafminer group. In 2020, $945 billion was lost due to cyber incidents, and another $145 billion was spent on cybersecurity. These costs have surged by more than 50% since 2018, when approximately $600 billion was allocated to mitigate cybercrime (B. Ballard, 2021).

In summary, APT attack detection requires specialized techniques that go beyond the traditional detection methods. These specialized techniques must address the unique challenges posed by APTs, including their stealth, persistence, targeted nature, sophistication, advanced evasion techniques, and focus on data exfiltration. Detecting APTs often involves leveraging advanced deep-learning methods, continuous monitoring, and the use of XAI to ensure transparency and trust in the detection process (Schwalbe and Finzel 2023).

1.1 Paper organization

Figure 2 provides the organization structure for this review. It outlines the main sections and subsections of the paper and provides a clear roadmap of the content and topics covered. In Sect. 2, we introduce the background of Advanced Persistent Threats (APTs), including the lifecycle and characteristics of these cybersecurity threats. Section 3 outlines the research methods, detailing the search strategy and eligibility criteria for selecting relevant literature. Section 4 delves into related work, discussing the classification of APT detection, various deep learning models used, and the limitations of existing detection methods. Section 5 presents the methodology, including a comparative analysis of the deep learning and XAI

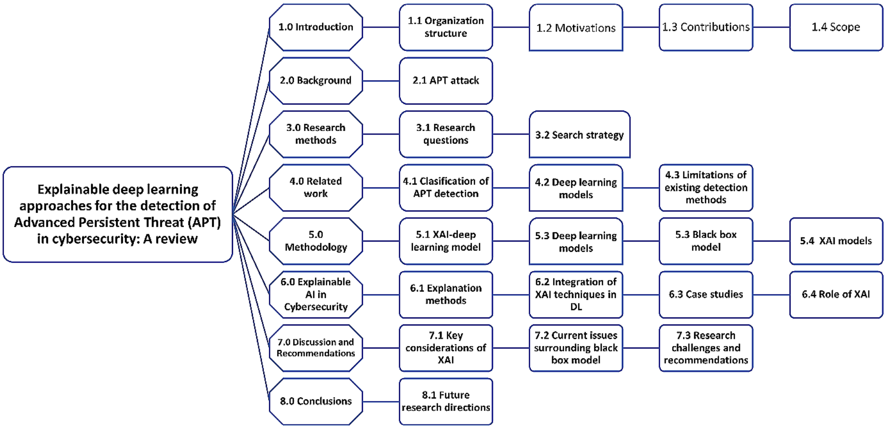


Fig. 2 The organization structure of the review

approaches, the need for XAI, and relevant case studies. Section 6 focuses on explainable AI in cybersecurity, explores explanation methods, integrates XAI techniques in deep learning, and addresses the current issues surrounding black box models. Section 7 discusses the findings, providing key considerations of XAI, challenges, and recommendations for future research. Finally, Sect. 8 concludes the review and suggests future research directions at the intersection of XAI and APT detection.

1.2 Motivation

The increasing sophistication of APT attacks and the limitations of existing IDS drive the key motivation for this review. Traditional IDS methods struggle with low detection accuracy, high false-positive rates, and the inability to detect unknown or early-stage attacks. For example, signature-based IDS can detect only known threats, making them ineffective against novel APTs (Sarker et al. 2024). Furthermore, these methods have difficulty detecting unknown or early-stage attacks because of their reliance on signature-based detection mechanisms, which fail to recognize novel threats.

Additionally, comprehensive studies detailing the current state of interpretability in published research and the application of state-of-the-art XAI models in the cybersecurity domain are lacking (Saeed and Omlin 2023). Without interpretability, cybersecurity experts cannot fully trust or understand the decisions made by AI models, which is crucial for accurately identifying false positives and negatives and for adapting to new attack vectors (Brown et al. 2022).

This review examines a critical gap in the literature on the implementation of deep learning techniques with XAI for APT detection. Table 1 provides a summary of various studies on APT detection methods, highlighting the objectives, motivations, and key findings of each study. This comparative analysis helps to identify gaps and challenges in existing methods, providing a foundation for enhancing state-of-the-art APT detection.

By addressing these gaps, our study aims to enhance the effectiveness and transparency of APT detection systems, thereby advancing the state-of-the-art in cybersecurity.

Table 1 Comparative literature studies on APT detection

Year	Author (s)	Objective	Motivation
2024	(Sarker et al. 2024)	Provide an extensive analysis of XAI based cybersecurity modeling in digital twin (DT) environments, focusing on the methods, taxonomy, challenges, and prospects.	The industrial transformation heavily relies on AI, including machine learning and data-driven technologies, to enable tasks such as self-monitoring, investigation, diagnosis, future prediction, and decision-making.
2023	(Salim et al. 2023)	Examined the different approaches used to detect APT attacks directed at the network system in terms of approach and assessment metrics	Traditional cyber defense mechanisms often fail to detect APTs due to their sophisticated nature and the rapid evolution of attack patterns. This detection failure underscores the need for more advanced and adaptive detection techniques.
2022	(Jabar and Mahinderjit Singh 2022)	Analyzed APT defense mechanisms were presented as a systematic literature review (SLR).	Limited resources, along with minimal storage capacity, limited computing power as well as specific energy resources, make smartphones lack built-in security and privacy protection.
2021	(Raju et al. 2021)	Discussed the feature representations, extraction techniques and ML models used in the reviewed studies.	To deal with limitations of dataset availability, lack of (1) devices, (2) security, and (3) software standards, model complexity, evasion methods, and others.
2020	(Stojanovic et al. 2020)	Analyzed datasets and descriptions utilized in the APT detection literature and feature extraction techniques employed in APT detection-related literature.	Advanced threats are very complex. This makes it hard to create models of attacks for training and testing.
2024	This review [2024]	This review aims to identify deep learning techniques integrated with XAI, assess the interpretability of existing research, and pinpoint the specific XAI methods utilized in the cybersecurity domain to enhance explainability and transparency.	To enhance the effectiveness and transparency of APT detection systems. As APTs become increasingly sophisticated, traditional detection methods often fall short in terms of accuracy, interpretability, and the ability to identify novel or early-stage attacks.

1.3 Contribution

In this review, we address the challenges of APT detection by integrating deep learning techniques with XAI methods. The following section discusses the key contributions of our research.

a. Our research expands the state-of-the-art in cybersecurity by providing robust, scalable, and interpretable detection systems that can effectively combat sophisticated APT attacks.
b. We evaluated robust APT detection frameworks that improve detection accuracy and scalability while providing clear, interpretable insights into the model's decision-making process. This approach ensures that cybersecurity experts can respond more effectively to threats and understand how and why specific decisions are made.
c. We highlight the limitations of existing detection systems, such as low detection accuracy and high false-positive rates. Our study demonstrates how integrating deep learning with XAI can address these challenges, paving the way for more robust and real-time detection capabilities.

d. We include case studies to demonstrate the application of XAI in combating APTs. These scenarios illustrate the potential impact of implementing XAI techniques in real-world settings, providing a comprehensive view of how these advanced methods can transform APT detection and response.
e. We examine various issues in the cybersecurity area and propose an improved method of APT detection on the basis of the specific attributes of such attacks. This tailored approach enhances detection capabilities and addresses the unique challenges posed by APTs.

In summary, our research provides significant contributions to the field of cybersecurity by integrating XAI APT detection systems. By addressing the current limitations and demonstrating the practical application of our methods, we pave the way for more robust, scalable, and transparent cybersecurity solutions. Our work enhances the overall resilience of network systems to APTs, ensuring better protection and response capabilities against sophisticated APT attacks.

1.4 Scope

The scope of this paper includes a comprehensive review of various deep learning techniques utilized for APT detection, the application of XAI methods to improve model interpretability and trustworthiness, and a critical analysis of existing approaches. By examining the intersection of deep learning and XAI, this paper aims to provide valuable insights and pave the way for more effective and reliable cybersecurity solutions. Our approach aligns with the need for explainability in AI models to ensure their practical application and trustworthiness.

Having established the importance of APT detection and the potential of XAI, we delve into the background study to provide a foundation for our research.

2 Background

In this section, we provide an overview of the background studies relevant to APT detection. Our focus is on the various methods that have been explored in the literature, such as deep learning and XAI approaches. This review aims to place our research within the wider scope of existing research and highlight the contributions we make to advancing the state-of-the-art in APT detection.

Deep learning approaches have gained significant attention for their ability to automatically learn complex patterns from large datasets. Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks have shown promising results when applied to APT detection. These models capture temporal and spatial dependencies in data, making them suitable for detecting sophisticated attack patterns. However, the black-box nature of deep learning models poses challenges in interpretability and transparency, which are critical for cybersecurity applications. To address the interpretability challenge, XAI techniques have been developed. XAI aims to make AI models' decision-making processes transparent and understandable to humans. Techniques such as SHAP and LIME help elucidate the contributions of individual features to a model's predictions, enhancing trust and facilitating collaboration between human experts and AI systems.

APTs pose a unique characteristic cyber-attack problem. They target specific victims and use tricks to hide. Skilled, well-funded attackers use various TTPs to evade detection (Sharma et al. 2023). APTs are managed by highly skilled and resourceful adversaries, often backed by nation-states or organized crime groups (Lemay et al. 2018). These threat actors have clear goals, such as espionage, sabotage, and damage, and spend much of their time succeeding (Ahmad et al. 2019). For example, the Lazarus Group established a method for compromising common software on the Internet with Trojans. These advanced attackers use social engineering techniques to inject malicious software into a target system. This allows them to attack the supply chain via untrustworthy sources (Villalón-Huerta et al. 2022).

The detection of APT attacks presents several distinct challenges compared with malware. The ways in which APT attack detection differs include the following:

- Stealth and persistence: APTs remain undetected for extended periods by blending with legitimate network traffic and activities (Bierwirth et al. 2024) (Moustafa & Slay, 2016). Unlike typical malware, which may execute its payload quickly, APTs focus on long-term espionage or data theft, making detection more difficult.
- Targeted nature: APTs are highly targeted and tailored to specific organizations or individuals, rendering generic security solutions less effective. Additionally, APT tactics, techniques, and procedures (TTPs) are continuously evolving (Ahmad et al. 2019) requiring deep learning models to be frequently retrained to detect new threats.
- Sophistication and complexity: APT attacks involve multiple stages, such as initial access, lateral movement (Fang et al., 2022), data exfiltration, and persistence. This multistep approach necessitates sophisticated detection mechanisms to identify and correlate various stages.
- Advanced evasion techniques: APTs use advanced evasion techniques, such as encryption, obfuscation, and polymorphism, to avoid detection by traditional methods (Shenderovitz & Nissim, 2024). They may also use legitimate system tools for malicious activities, complicating detection.
- Focus on data exfiltration: While malware might aim to cause immediate damage or disruption, APTs primarily aim to steal sensitive information, posing significant detection challenges (Sharma et al. 2023a).
- Nation-State Involvement: Many APTs are linked to nation-state actors conducting espionage or intelligence gathering, leveraging vast resources and advanced capabilities (Holt et al., 2023) (Bierwirth et al. 2024).
- Mitigation Strategies: Addressing the threat of APTs requires a multilayered approach (Mohamed, 2023) that includes proactive network monitoring, continuous vulnerability assessments, user awareness training, and robust incident response plans.

2.1 APT attack

APTs follow a well-defined lifecycle that consists of several stages. Initially, attackers infiltrate the target system using methods such as phishing, spear-phishing, or exploiting vulnerabilities in web applications. Once inside, they establish a foothold by installing malware to maintain persistence within the system. Subsequently, they escalate privileges by exploiting system vulnerabilities to gain deeper access. The attackers then perform internal reconnaissance, moving laterally within the network to map infrastructure and identify critical assets. In the next stage, attackers extract sensitive data and send it to the external servers they control. Finally, attackers maintain persistence using advanced evasion techniques to remain undetected and retain long-term access to compromised systems (Bierwirth et al. 2024).

APTs are characterized by their sophisticated and prolonged attack campaigns, which follow a well-defined lifecycle. The APT attack life cycle can be divided into six phases, intelligence gathering, point of entry, command and control (C&C), lateral movement, data of interest and the external server. The initial phase of an APT attack is reconnaissance or intelligence gathering, in which the attackers gather information about their target organization such as network infrastructure and identify potential vulnerabilities. This phase involves passive information gathering techniques and social media analysis (Bodstrom & Hamalainen, 2019). Once the attackers have identified potential entry points, they move to the establish phase, where they prepare customer malware to exploit the target environment. If successful, the attackers establish a foothold in the compromised systems, installing backdoors and remote access tools (RATs) to maintain persistence. From this point, the APT actors proceed to the command and control (C&C) phase, establishing secure communication channels with their malicious infrastructure to receive updates and exfiltrate data. With a persistent presence in the target network, the attackers move to the lateral movement phase, where they seek to escalate privileges and gain access to additional systems and resources. The final stages of the APT lifecycle involve the actual achievement of the attackers' objectives, such as data of interest, data exfiltration, sabotage, or establishing long-term access for future attacks (Chen et al., 2018).

In summary, APTs represent a sophisticated and persistent threat to organizations worldwide. Their ability to remain undetected and adapt to new defenses makes them particularly challenging to address. Recent high-profile cases, such as the SolarWinds (Alkhadra et al. 2021) and Hafnium attacks, highlight the critical need for advanced detection methods. This paper aims to explore the integration of deep learning techniques with Explainable AI (XAI) to enhance APT detection and provide more transparent, interpretable solutions for cybersecurity experts.

3 Research methods

In this section, we outline the research questions and search strategy that guided our study. Next, we explain the research process.

3.1 Research questions

We established a set of research questions to address important aspects of APT detection. These questions will guide the development of an effective detection system. They are designed to explore the performance, limitations, and potential enhancements of APT detection systems. Table 2 provides description of the research questions as follows:

This section outlines the key research questions that drive our investigation into APT detection. By addressing these questions, we aim to uncover the strengths and weaknesses

Table 2 Research questions

No.	Research question	Description
RQ1	How do existing detection systems identify APT, and what are their limitations?	Investigate the methodologies and mechanisms used by current APT detection systems, analyzing their strengths and weaknesses.
RQ2	How can XAI be integrated into APT detection systems to enhance transparency and trustworthiness?	Explore the integration of XAI into APT detection frameworks to improve transparency, accountability, and user trust.
RQ3	What XAI techniques have been proposed to detect APTs?	Identify the categories of XAI techniques applied in APT detection.
RQ4	How can XAI techniques improve the interpretability and effectiveness of deep-learning models for APT detection?	Examine how various XAI methods can be employed to make deep learning models more interpretable and effective in detecting APTs.
RQ5	What are the challenges in applying XAI for APT detection?	Identify the key obstacles and limitations in implementing XAI techniques of APT detection.

of current systems, explore the benefits of integrating XAI, and identify the unique challenges in applying these advanced techniques.

3.2 Search strategy and eligibility criteria

In this review, we conducted a rigorous search strategy and defined clear eligibility criteria for the selection of relevant papers. We searched multiple academic databases, including IEEE Xplore, ACM Digital Library, ScienceDirect, Springer Link, Web of Science, and Google Scholar. These databases were chosen for their relevance to research in computer science, cybersecurity, and artificial intelligence. We used specific search terms to find studies related to APT detection mechanisms. Boolean expressions such as ("Advanced Persistent Threat" OR "APT") AND ("Deep Learning AND "Cybersecurity") AND ("Explainable Artificial Intelligence" OR "XAI") were used to combine search terms.

Our review aimed to include papers presenting novel methods, particularly those involving deep learning-based APT detection, the application of deep learning, and XAI techniques, along with empirical results. The inclusion criteria required papers to be published in English in peer-reviewed journals, conference proceedings, or book chapters from 2018 to 2024. We excluded papers without abstracts, without access to full content, or not written in English.

We reviewed selected articles that met our inclusion criteria, focusing on those that provided insights into the integration of XAI with APT detection. Through this process, we collected 100 articles deemed the most relevant to our study. Figure 3 provides an overview of our search strategy and eligibility criteria. To refine the search results and relevant papers, we applied the following inclusion criteria:

a. The paper must be written in English and published in peer-reviewed journals, conference proceedings, or book chapters.
b. The study must focus on applying deep learning techniques specifically for detecting APT attacks.
c. The study must involve applying XAI methods in cybersecurity, specifically for APT detection.

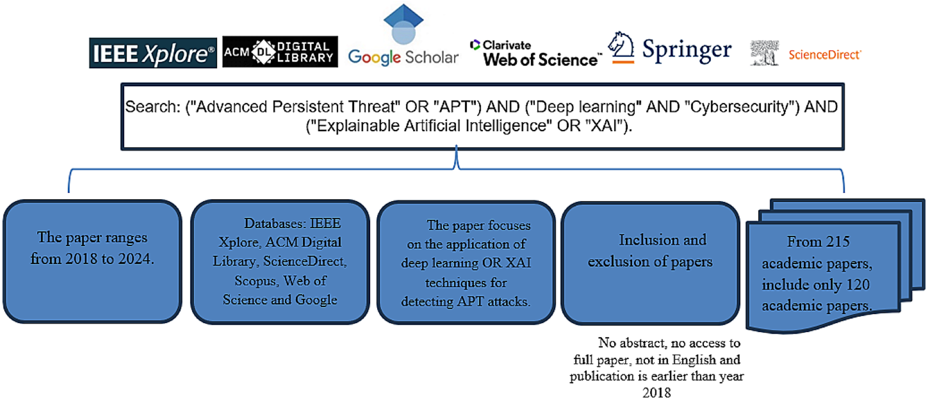


Fig. 3 Overview of the search strategy and eligibility criteria

d. The study must provide empirical results, such as experiments or case studies, which demonstrate the effectiveness of the proposed approach in real-world or simulated scenarios.
e. The papers must have been published between 2018 to 2024, reflecting recent advancements in deep learning and XAI research.

This section outlines the rigorous search strategy and well-defined eligibility criteria used to select relevant papers for this review. The search was conducted across several major academic databases using specific terms related to APT detection, deep learning, and XAI.

4 Related work

In this section, we review the existing literature on APT detection methods, focusing on both traditional and advanced approaches. We organize the discussion to provide a clearer understanding of the strengths and weaknesses of each approach relative to our study. Current IDS and security measures face significant limitations when dealing with APTs.

4.1 The classification of APT detection approaches

Classifying APT detection approaches on the basis of data sources such as Network Traffic Analysis, Host-based Analysis, and Log and Event Analysis allows for a more targeted and efficient detection strategy. A study by (Do Xuan et al. 2020) analyzed network traffic into IP-based network flows, reconstructed IP information from these flows, and used deep learning models to extract features to distinguish APT attack IPs from other IPs. They introduced a combined deep learning model using Bidirectional Long Short-Term Memory (BiLSTM) and Graph Convolutional Networks (GCN). Meanwhile, host-based analysis focuses on the data and activities occurring on individual hosts or endpoints within a network.(Chen et al. 2024) proposed a hybrid Network Intrusion Detection System (NIDS) that combines host-based intrusion detection (HIDS) and network-based detection to enhance the detection of APTs and other network intrusions. Log and Event Analysis involves collecting and analyzing logs and events generated by various systems, applications, and devices within an organization. (Wang et al. 2022) presented a novel method for reconstructing APTs in large-scale networks to improve attack forensics and traceability. The proposed method addresses these issues with a low transmission cost and does not require raw data from terminal devices, making it suitable for extensive networks with numerous terminal devices. Classifying APT detection approaches based on explainability involves distinguishing between black-box models and XAI models. The black-box models will be explained in Sect. 5.4. XAI models offer transparent and interpretable insights into the detection process. These models aim to maintain a high level of accuracy while ensuring that the reasoning behind their predictions is clear and understandable to cybersecurity experts. This transparency increases trust in the detection system and facilitates a faster and more effective response to identified threats.

4.2 Deep learning models

Machine learning (ML) is a key component of data science, and is important in terms of its flexibility, scalability, and adaptability to new challenges. This article explores ML applications in cybersecurity, including phishing detection, network intrusion detection, spam detection in social networks, smart meter energy consumption profiling, and security concerns inherent in ML techniques. To understand this, we have added Fig. 4, which shows taxonomies based on specific themes and deep learning (DL) methods. This structured approach will not only add more organization to our review but also increase its reference value by providing a clearer framework. This study emphasizes the methodology of collecting large datasets, extracting relevant features, and training ML models using supervised learning algorithms to achieve high accuracy and low false positive rates.

(Manoharan et al. 2023) addressed the challenge of detecting insider threats, which pose significant cybersecurity risks. This study evaluated various supervised machine learning algorithms on a balanced dataset using the same feature extraction method and investigated the impact of hyperparameter tuning and different conditions on imbalanced datasets. Using the publicly available CERT r4.2 dataset, the results showed that Random Forest achieved the best accuracy and F1-score of 95.9%, outperforming existing methods such as the DNN, LSTM Autoencoder, and User Behavior Analysis. (Raju et al. 2021) explored various ML models, including decision trees and support vector machines, to detect APT activities. Although these models offered improved detection capabilities over traditional methods, they still struggled with interpretability. Cybersecurity experts often find it difficult to understand the reasoning behind ML predictions, which hampers their trust and usability. (Stojanović et al. 2020) proposed the use of ensemble methods to combine multiple ML models, thereby enhancing detection accuracy. However, this approach increases computational complexity and may not be suitable for real-time applications. Our study leverages the strengths of ML while addressing its weaknesses by using deep learning and XAI to provide more interpretable and scalable solutions.

Deep learning is a widely used technique for APT detection, offering several advantages over traditional machine learning techniques. These advantages include the ability to process and analyze unstructured data, such as text, images, and network traffic, as well as high-dimensional data, which are common in cybersecurity domains. Unlike traditional methods, which often rely on predefined rules and signatures, deep learning models can automatically learn complex patterns and features from large datasets. This capability allows for more

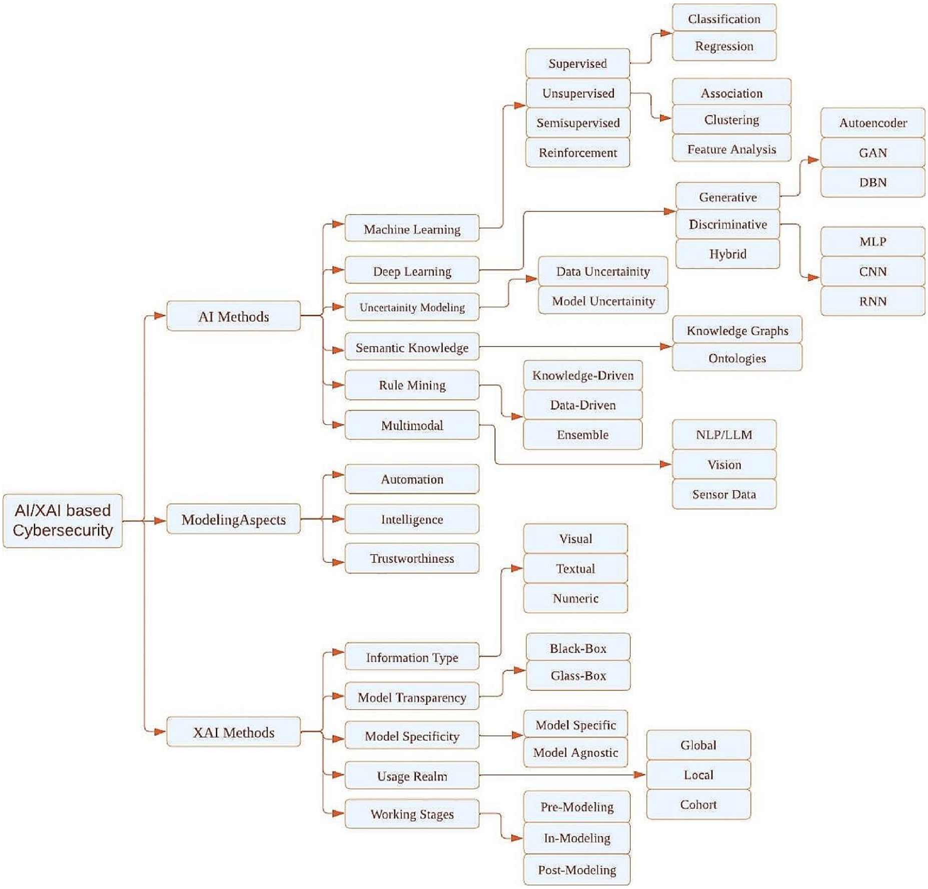


Fig. 4 A taxonomy of AI/XAI based methods for cybersecurity modeling. Adapted this figure from (Sarker et al. 2024)

accurate and robust detection of sophisticated cyber threats, making deep learning an essential tool in the evolving landscape of cybersecurity (Alzubaidi et al. 2021).

(Mittal et al. 2023) conducted a systematic review on deep learning for detecting Distributed Denial of Service (DDoS) attacks, analyzing literature from multiple sources. They categorize findings into five key areas: deep learning approaches, methodologies, datasets, preprocessing strategies, and research gaps. This study evaluates existing methods, highlights strengths and weaknesses, and identifies gaps in current research, suggesting future directions. This review offers a comprehensive overview, is organized for clarity, and emphasizes the relevance of deep learning in addressing the evolving threat of DDoS attacks. Despite its thoroughness, the study's reliance on existing literature and the rapid evolution of attack strategies may limit its timeliness and practical application.

4.2.1 Convolutional neural networks (CNNs)

Among the various types of deep learning techniques, Convolutional Neural Networks (CNNs) excel at handling high-dimensional data and automatically learn and extract relevant spatial features from raw data. This capability makes them particularly effective in identifying complex patterns associated with APT activities. In computer vision, researchers use CNNs to recognize APTs by extracting features from APT data (Teuwen and Moriakov 2020). The CNN architecture includes five block layers: convolutional layers, pooling layers, channel max-pooling layers, rectified linear unit (ReLU) layers, fully connected layers, and a SoftMax loss function (Alzubaidi et al. 2021). However, CNNs have limitations in converting one-dimensional traffic flows into two-dimensional flows without considering the spatial correlations between the traffic flows. (Alzubaidi et al. 2021)

(Jayapradha et al. 2024) proposed an innovative intrusion detection system (IDS) that leverages deep learning algorithms to enhance phishing detection. Specifically, CNNs automatically extract sophisticated features from raw input data, whereas RNNs effectively model sequential data to recognize phishing patterns over time. This IDS adapts in real-time to new phishing variants through back propagation-based model optimization, significantly improving detection accuracy compared with traditional rule-based methods. By utilizing the KDD-CUP99 dataset for training, the study demonstrates a robust defense against evolving phishing threats. Consequently, the system enables a proactive incident response and strengthens overall network security. Despite potential challenges in terms of computational complexity and data dependency, the proposed system offers substantial improvements in protecting networks and users from sophisticated phishing attacks.

(Patel et al. 2024) proposed an advanced malware detection system that leverages RNNs and CNNs to enhance cybersecurity vigilance. The study highlights a pioneering capability by extending detection to unconventional formats such as GIFs and images, addressing emerging threats that exploit multimedia channels. This approach demonstrates the adaptability and comprehensiveness of the deep learning-based system. Traditional cybersecurity methods struggle to keep pace with dynamic cyber adversaries, but this research leverages deep learning's ability to recognize complex patterns within vast datasets. By integrating RNNs and CNNs, this study aims to improve the accuracy of threat identification and the system's resilience against emerging and previously unseen cyber threats. This study focuses on broadening the detection scope to multimedia formats and enhancing the system's adaptability and reliability. The strengths of this study lie in its innovative approach and comprehensive coverage of diverse data formats, whereas its weaknesses include potential computational complexity and reliance on high-quality datasets for training.

(Tadesse and Choi 2024) proposed a novel intrusion detection system (IDS) that converts raw datasets into image datasets using the Short-Term Fourier Transform (STFT) to enhance pattern recognition. This system uses a lightweight convolutional neural network (CNN) to classify denial of service (DoS) and distributed denial of service (DDoS) attacks. Evaluated on both modern (CSE-CIC-IDS2018) and legacy (NSLKDD) datasets, the proposed methods achieve high accuracy and low false alarm rates, demonstrating high specificity and sensitivity. The study highlights the model's excellent generalizability and avoidance of overfitting across different datasets, emphasizing the effectiveness of the dataset conversion methodology.

(Najar & S., 2024) proposed an innovative feature selection approach to develop a robust and reliable intrusion detection system (IDS) for detecting and classifying Distributed Denial-of-Service (DDoS) attack types. Using the CICDDoS2019 benchmark dataset, the model demonstrates high performance, achieving 96.82% accuracy, 96.82% recall, 96.76% precision, and a 96.50% F1 score. It also provides rapid prediction times, identifies attacks in just 0.189 milliseconds, and outperforms existing methods and baseline models. The contributions include a novel feature selection approach, effective preprocessing techniques, and comprehensive evaluation using benchmark datasets. The strengths of this study are its high detection accuracy, rapid response, balanced performance metrics, and innovative methodology. However, weaknesses include challenges with imbalanced data handling, computational complexity, dependency on dataset quality, generalizability to other attack types, and implementation challenges.

(Korium et al. 2024) proposed an advanced IDS tailored for the Internet of Vehicles environment, leveraging CNNs to detect both traditional and new cyber-attacks effectively. The novel network architecture of the model at the data processing layer and the use of the synthetic minority oversampling technique significantly enhanced the detection accuracy and speed, achieving a notable 94% detection rate using the AWID dataset. The strengths of this study are its high detection accuracy, improved data processing through synthetic oversampling, innovative network architecture, and comprehensive evaluation with the AWID dataset. The model's performance is heavily dependent on the AWID dataset, and its scalability may be limited by the computational complexity introduced by CNNs and data processing techniques.

(Sun 2024) proposed a novel intrusion detection system (IDS) that combines data preprocessing with four deep neural network models: Convolutional Neural Networks (CNN), Bi-directional Long Short-Term Memory (BiLSTM), Bidirectional Gate Recurrent Unit (BiGRU), and Attention mechanism to identify network attacks accurately. Evaluated using the NSL-KDD dataset, the models utilize preprocessing techniques and particle swarm optimization for feature selection, with hyperparameter tuning via the BO-TPE algorithm. The study's contributions include a comprehensive IDS framework, the introduction of multiple DNN architectures, and effective feature selection and class imbalance handling, resulting in high detection accuracy rates of 0.999158 in binary classification and 0.999091 in multiclass classification. The strengths of this study include its thorough approach, high accuracy, and advanced optimization techniques, whereas its weaknesses include dataset dependency, computational complexity, generalizability concerns, implementation challenges, and the need for further real-world application exploration.

(Yin et al. 2024) introduced a novel multi-scale Convolutional Neural Networks (CNN) and bidirectional Long-short Term Memory (bi-LSTM) arbitration dense network model (MSCBL-ADN) for detecting LDDoS attacks effectively with limited datasets and reduced time consumption. The MSCBL-ADN model integrates a CNN for spatial feature extraction, bi-LSTM for temporal relationship extraction, an arbitration network to re-weigh feature importance, and a 2-block dense connection network for final classification. The experimental results on the ISCX-2016-SlowDos dataset demonstrate that the MSCBL-ADN model significantly improves detection accuracy and time performance compared to state-of-the-art models.

(Ersavas et al. 2024) explored the potential of Convolutional Neural Networks (CNNs) beyond traditional image processing, highlighting their ability to analyze high-dimensional datasets by transforming them into pseudo-images. Despite the rise of newer architectures such as Transformers, CNNs remain crucial in various applications, including Generative AI. The authors introduce DeepMapper, a pipeline that enables the analysis of complex datasets without intermediate filtering or dimension reduction, thus preserving data integrity and detecting small variations typically considered noise. They demonstrated that DeepMapper can efficiently and accurately identify subtle perturbations in large datasets with numerous features, highlighting its superiority in speed and comparable accuracy to existing methods.

Mendonça et al. (2023) proposed a new IDS using a hierarchical tree-CNN algorithm with soft-root-sign (SRS) activation (Daoud et al. 2023) to detect attacks in infiltrations, Distributed Denial-of-Service (DDoS), brute force, and web attacks. The authors reported that their proposed system could detect DDoS attacks with high accuracy and a reduced execution time of approximately 36%. Furthermore, the results showed a significant increase in the average detection accuracy of 98% when all the analyzed attacks were considered. This indicates that Tree-CNN performs better because it is less complex, requires less processing time, and consumes fewer computing resources than other current machine-learning-based IDSs do.

On the other hand, Zhu & Zu, 2022) implemented a fully convolutional neural network (FCNN) classifier architecture that substitutes the linear layers and correlated activation functions from prevalent CNN classifiers. They trained the FCNN classifier using SoftMax loss. The main benefits of this architecture are its simplicity and flexibility. However, it does not consider channel information. They can use SoftMax loss directly to train the network by reconstructing the number of output channels. The predefined best distribution (POD loss) of the latent features was used to improve the recognition rate performance with SoftMax loss.

Network Intrusion Detection Systems (NIDS) are essential for detecting malicious activities in modern networks. However, class imbalance in intrusion detection datasets hinders the performance of classifiers in minority classes. To address this, (Zhang et al. 2020) proposed a novel class imbalance processing technique called the SGM, which combines the Synthetic Minority Over-Sampling Technique (SMOTE) and under-sampling using Gaussian Mixture Model (GMM). They developed a flow-based intrusion detection model, SGM-CNN, which integrates imbalance processing with a convolutional neural network (CNN). They evaluated the performance on the UNSW-NB15 and CICIDS2017 datasets, and reported high detection rates of 99.74% for binary classification, 96.54% for multi-class classification for the UNSW-NB15 dataset, and 99.85% for 15-class classification for the CICIDS2017 dataset. The authors claimed that SGM-CNN effectively addresses the challenge of imbalanced intrusion detection and outperforms the existing state-of-the-art methods.

(Tian, 2020) proposed a combination approach, the CNN algorithm, to learn the deep features of an image using CNNs and RNNs in parallel. The authors subsequently created a ShortCut3-ResNet residual module. In their study, they demonstrated that the convolutional neural network algorithm can identify various features of images, optimize the accuracy of feature extraction, and improve the ability of the convolutional neural network to recognize images.

CNNs have shown significant promise in enhancing cybersecurity measures through various innovative approaches. These studies highlight the adaptability of CNNs in recognizing complex patterns, handling diverse data formats, and integrating with other deep learning models to improve detection accuracy and resilience against sophisticated cyber threats. Despite challenges such as computational complexity and data dependency, CNN-based systems offer substantial improvements in protecting networks and users from evolving threats.

4.2.2 Recurrent neural networks (RNNs)

Recurrent Neural Networks (RNNs) are specifically designed to handle sequential data, making them particularly suitable for analyzing time-series data in network traffic. By incorporating internal memory states, RNNs can effectively process input sequences of varying lengths, capturing temporal dependencies within the data (DiPietro and Hager 2020). This capability allows RNNs and their advanced variants, such as Long Short-Term Memory (LSTM) networks, to model the sequential nature of APT attack campaigns (Yuan et al. 2017). RNNs have proven effective for APT detection because they can learn from event sequences over time and predict future actions based on past observations. This makes RNNs an invaluable tool for identifying and mitigating sophisticated cyber threats (Galli et al. 2024).

In the current landscape of complex cyber threats, network security is of utmost importance. (Kumaresan et al. 2024) evaluated the efficacy of Recurrent Neural Networks (RNNs) in network anomaly detection, comparing them with traditional methods such as statistical techniques and simple neural networks. Using an extensive dataset of normal and malicious network traffic, the authors demonstrate the potential of RNNs to detect anomalies by utilizing sequential dependencies in the data. They investigated various RNN architectures, hyperparameter settings, and feature representations to improve detection performance. The paper also addresses challenges such as model interpretability, scalability, and computational resource demands, and proposes ways to increase the resilience of RNN-based systems against malicious interference.

(Yang et al. 2024) introduced the Hypergraph Recurrent Neural Network (HRNN), a novel intrusion detection method that leverages hypergraph structures and recurrent networks. The HRNN represents flow data as hypergraph structures to enhance information representation and incorporates a recurrent module to extract temporal features. This design integrates rich spatial and temporal semantics, significantly improving anomaly detection capabilities. Evaluations on several publicly available datasets demonstrate that the HRNN outperforms other state-of-the-art methods, demonstrating its superior performance in detecting network anomalies. However, the increased complexity due to the use of hypergraph structures and recurrent networks may impact computational efficiency and scalability. Furthermore, the performance of the HRNN may heavily depend on the quality and characteristics of the datasets used for training and evaluation, potentially requiring fine-tuning for different data forms and limiting its transferability. Integrating HRNN with existing network infrastructure might also present technical challenges.

(Saravanan et al. 2023) proposed a Blockchain-based African Buffalo (BbAB) scheme with a Recurrent Neural Network (RNN) to enhance an IDS. The method encrypts normal and malware user datasets using Identity Based Encryption (IBE) and securely stores them in a blockchain within a cloud environment. The RNN detects intrusions, using African buffalo optimization for continuous monitoring. The approach achieves 99.87% accuracy and 99.92% recall, demonstrating robust detection capabilities. While the method improves security and monitoring, it presents challenges such as high computational complexity and significant resource requirements. Overall, the model enhances IDS effectiveness in cloud environments.

(Pahuja and Ojha 2024) addressed the growing threat of network attacks by deploying deep learning techniques, including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), to detect and mitigate Denial-of-Service (DoS) attacks. These models analyze sequential time-series data to identify patterns linked to DoS attacks. Among the techniques evaluated, LSTM achieved the highest accuracy of 92.3%, demonstrating superior capability in classifying attack traffic. This approach aims to reduce system unavailability and potential losses caused by DoS attacks, although it faces challenges in computational complexity and scalability.

(Sakthipriya et al. 2024) proposed enhancing IoT network security by reducing data dimensionality for efficient attack classification on memory-constrained devices. They used a Conditional Adversarial Auto Encoder (CAAE) to generate realistic botnet traffic and extract deep features, combined with a Dilated and Cascaded Recurrent Neural Network (DC-RNN) for accurate classification. The model achieves 96% accuracy, 98% precision, 97% F1-score, and 96% recall. This approach addresses deep learning implementation challenges in IoT environments, outperforming conventional methods. However, it faces issues with computational complexity, data dependency, scalability, generalizability, and implementation challenges in existing IoT infrastructures.

(Hasan et al. 2023) proposed an effective identification model for APT attacks by boosting-based machine learning methods combined with XAI to enhance prediction and provide actionable insights. The model, which achieves a high weighted F1 score of 0.97 with XGBoost, effectively predicts APT attacks and utilizes SHAP to make predictions understandable and actionable for cybersecurity stakeholders. While the approach demonstrates high detection accuracy and offers a promising framework for future research, it faces challenges such as computational complexity, data dependency, generalizability, implementation difficulties, and resource intensity, potentially impacting scalability and practical application in resource-constrained environments.

Furthermore, Lo et al. (2023) introduced the XG-BoT GNN and used GNNExplainer on a botnet graph dataset. Their model demonstrated impressive precision between 99.23% and 99.63%. The effectiveness of this approach relies heavily on the quality of the input data.

AlDahoul et al. (2021) developed a fusion model that combined two DNNs, trained using class-weight optimization. The main idea is to learn complex patterns from rare anomalies in the traffic data. The authors used Adam optimization algorithms with classes to train the DNNs. Their results showed that their method could achieve a higher accuracy in terms of the Fβ score and the false alarm rate when their proposed model compared to conventional single DNNs. Furthermore, their experiment used the ZYELL real-world datasetand yielded promising results.

These studies demonstrate the versatility and effectiveness of RNNs and their variants in enhancing cybersecurity measures, particularly in detecting and mitigating sophisticated cyber threats. Despite challenges such as computational complexity and data dependency, RNN-based systems offer substantial improvements in network security.

4.2.3 Autoencoders

Autoencoders (AEs) are used for anomaly detection to represent normal behavior and subsequently identify deviations from this learned behavior. These neural network models are trained on datasets comprising normal activity, enabling them to compress and reconstruct input data accurately. When presented with anomalous data, autoencoders produce higher reconstruction errors, thus flagging the anomalies. They have shown significant promise in detecting APTs by highlighting unusual activities within the network traffic, such as unexpected patterns that differ from established norms, thereby providing a robust mechanism for identifying potential security breaches.

(Yashwanth et al. 2024) proposed a novel approach by combining Auto-encoders with Multi-Layer Perceptron (MLP). The study evaluates three algorithms: Auto-encoders, Auto-encoders with MLP, and CNNs, and the results demonstrate their effectiveness in detecting network intrusions. This study highlights the significance of using diverse machine learning techniques for effective anomaly detection, pattern recognition in complex network traffic, and handling imbalanced data, although challenges in computational complexity, resource requirements, and implementation in existing IDS infrastructure exist. Table 3 summarizes the deep learning-based methods used in cybersecurity.

4.3 Limitations of existing detection methods

The application of deep learning for APT detection presents several challenges. Traditional IDSs typically employ predefined signatures of known threats, which fail to detect new, unknown, or evolving APT tactics that do not match existing signatures. Anomalybased detection systems generate numerous false positives by flagging benign activities as potential threats, leading to alert fatigue and overwhelming cybersecurity experts, thereby reducing the overall effectiveness of the security operations center (SOC). Detecting early-stage attacks, such as initial compromise and establishing a foothold, is difficult because these activities often involve subtle actions that are difficult to distinguish from normal behavior. Traditional methods lack the ability to detect early indicators of compromise. APTs employ sophisticated evasion techniques, including encryption, polymorphism, and the use of legitimate system tools for malicious purposes, making traditional IDSs challenging. Furthermore, conventional methods often lack the ability to contextualize alerts within a broader network environment, failing to correlate seemingly unrelated events that together indicate an ongoing APT attack.

The limitations of the existing detection methods emphasize the need for advanced techniques that can effectively identify and mitigate APTs. Deep learning is a promising approach because of its ability to analyze large volumes of data and identify complex patterns indicative of APT activities. For example, using Convolutional Neural Networks (CNNs) to detect anomalies in network traffic data can uncover subtle changes indicative of a potential threat that traditional methods might miss. XAI can highlight the specific features of the data that led to this detection. However, the black-box nature of deep-learning models poses challenges in terms of interpretability and trust, making it difficult for cybersecurity experts to understand and act upon the model's predictions (Hassija et al., 2024). XAI enhances the transparency and interpretability of deep learning models, enabling cybersecurity experts to

Table 3 Summary of deep learning-based methods used in cybersecurity

Author (s)	Application	Model	Datasets	Metrics	Results
Jayapradha et al. 2024	Phishing detection	CNNs and RNNs	KDD-CUP99	Accuracy Precision F1-score Recall	97.2 99 99 99
Tadesse and Choi 2024	Intrusion detection (DoS/DDoS)	CNN, Short-Term Fourier Transform (STFT)	CSE-CIC-IDS2018, NSL-KDD	Accuracy Precision	99.37 99.37
Najar & S., 2024	DDoS attack detection	Feature selection approach	CICDDoS2019	Accuracy	96.82
Korium et al. 2024	IDS for IoV	CNNs, synthetic minority oversampling	AWID	Accuracy	94
Sun 2024	Network attack detection	CNN, BiLSTM, BiGRU, Attention mechanism, particle swarm optimization, BO-TPE algorithm	NSL-KDD	Accuracy	99.9158 (binary), 99.9091 (multi-class)
Yashwanth et al. 2024	Network intrusion detection	Autoencoders with Multi-Layer Perceptron (MLP)	NSL-KDD	Accuracy Precision	98 96.25
Sakthipriya et al. 2024	IoT network security	Conditional Adversarial Auto Encoder (CAAE), DC-RNN	N-BaIoT Dataset	Accuracy Precision F1-score Recall	96 98 97 96
Saravanan et al. 2023	IDS	Blockchain-based African Buffalo (BbAB) scheme with RNN	Normal and malware user datasets	Accuracy	99.87
(Zhu & Zu, 2022)	Classifier architecture	Fully convolutional neural network (FCNN)	CIFAR10, CIFAR100, Tiny ImageNet and FaceScrub dataset	Accuracy	84.24–93.43
(Lee et al., 2022)	Network intrusion detection	AE and DNN	Two individual datasets	Accuracy	97.28 89.94
(Khan et al. 2022)	Network intrusion detection	LSTM, AE	gas pipeline, and UNSWNB-15 datasets	Accuracy	97.95 and 97.62
(AlDahoul et al. 2021)	Detected anomalies in traffic data	Fusion model, two DNNs, Adam optimization algorithms	ZYELL real-world dataset	Average macro Fβ score and false alarm rate	17 5.3
(Mendonça et al., 2021)	IDS	Tree-CNN algorithm with soft-root-sign (SRS) activation	CICIDS2017	Accuracy	98

gain insight into the model's decision-making process. This not only improves the effectiveness of APT detection, but also fosters trust and reliability in automated security systems.

By focusing on the unique characteristics of APTs and addressing the specific limitations of current methods, this review aims to improve state-of-the-art APT detection and provide practical solutions for enhancing cybersecurity defenses. Cybersecurity experts must understand the explanations behind the model's predictions.

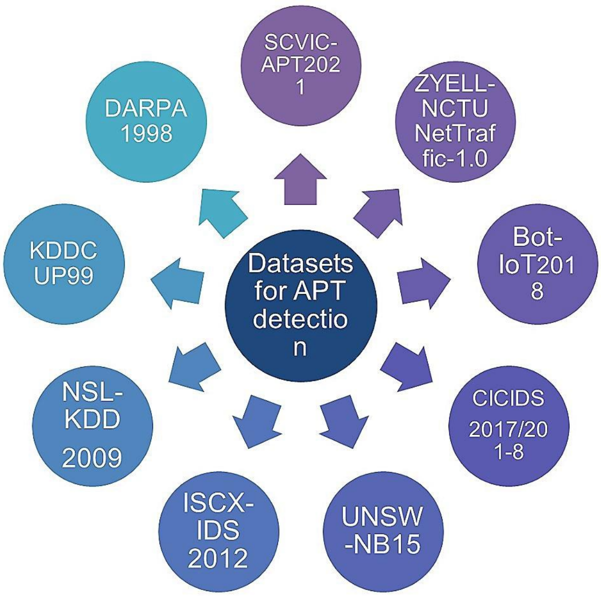
4.4 Datasets

In this section, we delve into datasets curated for APT detection. Figure 5 shows an overview of the most widely used methods in this field. In cybersecurity, benchmark datasets play a critical role in APT attacks (Agrawal et al. 2024).

We categorize the datasets based on their characteristics, such as the type of data, network traffic, system logs, presence of labeled attack scenarios, and extent of real-world applicability. This analysis helps to understand the strengths and limitations of each dataset, guiding researchers in selecting the most appropriate data for their studies. However, publicly available datasets that capture the behavior of APT attacks are lacking (Khraisat et al. 2019). This limitation hinders the development of effective APT detection models. Some existing datasets for analyzing APT attacks, such as DARPA1998, NSK-KDD 2009, UNSW-NB15, CICIDS2017, and ZYELL, focus on general network intrusion detection. However, they do not specifically target APT attacks.

DARPA1998 is the first dataset collection of attack traces from the internet. MIT Lincoln compiled and distributed it under DARPA and ARFL to evaluate network intrusion detection. It has seven weeks of training data and two weeks of testing data containing 38 attacks from four diverse groups: DoS, U2R, R2L, and probe (Homoliak et al. 2020). Another widely used dataset is the NSL-KDD (2009), which is a revised version of the KDD99 dataset. This revision reduces classifier bias and provides better detection rates (V C et al., 2023). Researchers commonly use the NSL-KDD for evaluation. This is an improved version of the original KDD-Cup99 benchmark. It considers 148,514 network traffic items across 41 features and five main attack types (Yang et al. 2021). The dataset includes denial-of-service attacks, remote access breaches, R2L, U2R, and probes from 77,054 benign samples (Barnard et al. 2022). The ISCX-IDS2012 dataset contains 2,381,532 data samples and is based on profiles that include intrusion details.

Fig. 5 Overview of datasets used in APT detection



The UNSW-NB15 dataset was created at the Australian Center for Cyber Security (ACCS) Cyber RangeLab, using the IXIA Perfect Storm tool. It exhibits both benign and malicious attacks. The dataset included 49 features, with a single-class label indicating the connection property of each data instance (Moustafa and Slay 2016). The dataset contains nine types of attacks: analysis, fuzzers, generic, exploits, DoS, backdoors, reconnaissance, worms, and shellcodes. Another dataset, CICIDS2017, was created based on a large-scale cybersecurity research project that collected information from more than three million computers globally beginning in April 2016. The CICIDS2017 comprises 78 features, 168,186 normal samples, and 2,180 attack samples (Patil et al. 2022).

The ZYELL dataset was generated to detect network anomalies using real-world network traffic data obtained from the ZYELL security system (L. Chen et al., 2021). The dataset comprises both benign and malicious network traffic with a proportion of anomalies of approximately 1%. It also targets two main types of attacks: probing and DoS (AlDahoul et al. 2021). It has 22 features, including connection duration and inbound/outbound traffic counts in bytes, and is stored as csv files (L. Chen et al., 2021). The dataset includes 9,241,463 training samples and 13,290,530 testing samples. However, creating an effective APT detection system using network datasets from different sources and organizations can be challenging because of privacy concerns and the lack of a universal dataset format.

In conclusion, a lack of comprehensive datasets remains a major challenge in the research and development of deep-learning models for APT detection (Karim et al. 2024). Collaborative efforts among researchers, industry partners, and government agencies are needed to create large-scale, diverse, and labeled datasets that can support the advancement of deep learning models in this domain. This will help the deep learning techniques in this field move forward. While previous studies have demonstrated the promise of deep learning and XAI in cybersecurity, there remains a need for integrated approaches that enhance both detection accuracy and interpretability.

5 Methodology

We outline the methodology used in our study on how XAI techniques are integrated with deep learning models. The methodology included the following key steps; data collection, data preprocessing, deep learning models, black box models, XAI models, and model explanations. Figure 6 provides an overview of the APT detection enhancement process, illustrating the various stages from data collection to model explanation. This framework aims to provide a clear understanding of the integration of different models and techniques for effective and concise APT detection.

5.1 Data collection

We gathered data from various sources, including network logs, system events, and user activities. The latest dataset from the Symantec Virtual Conference (SVC) 2021, SCVIC-APT-2021 (Liu et al. 2022) were utilized due to their comprehensive logs of simulated APT attacks and normal network traffic.

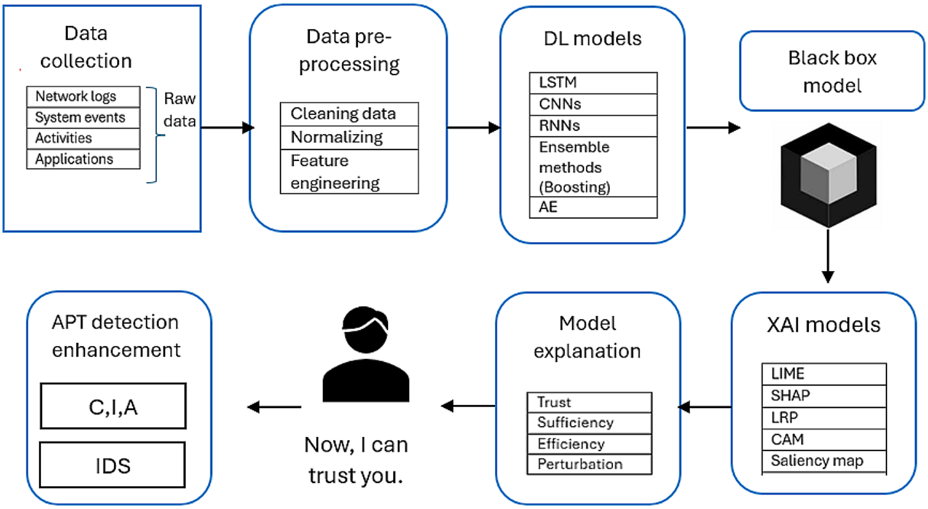


Fig. 6 Proposed XAI pipeline for APT detection

5.2 Data preprocessing

To ensure the quality and consistency of the data, the following preprocessing steps were performed:

1. **Data cleaning** involves identifying and removing irrelevant or redundant data points to improve the quality of the dataset (Sakthipriya et al. 2024) (Ridzuan and Zainon 2019). Irrelevant data points, such as incomplete logs or noise generated by benign activities, do not contribute to APT detection. Redundant data points are duplicate records that can skew the analysis.
2. **Normalization** ensures that data values are standardized to a common scale, which is crucial for the performance of machine learning models (Davis et al. 2020). Different features in the dataset may have varying scales, and normalization helps bring them to a comparable range (Siddiqi and Pak 2021).
3. **Feature engineering** involves identifying and extracting relevant features that can distinguish between normal and malicious activities (Abbas et al. 2023). This process enhances the model's ability to detect APTs by providing it with informative and discriminative attributes (Abu Bakar et al. 2023).

5.3 Deep learning models

Deep learning models have significantly advanced the field of cybersecurity, particularly in detecting advanced persistent threats (APTs). We utilized several deep learning models, each suited to different types of data:

Convolutional Neural Networks (CNNs) are widely employed for their ability to automatically learn and extract spatial features from raw data, making them effective for high-dimensional data (Taye 2023). CNNs apply convolutional layers to input data, use filters to detect spatial hierarchies and identify various patterns and structures, which is particularly useful for datasets such as images or network traffic logs (Ersavas et al. 2024). CNNs automatically extract spatial features through layers of convolution and pooling. This hierarchical feature extraction process enables CNNs to construct increasingly abstract representations of the input data, from low-level details to high-level concepts (Geng and Niu 2024). Leveraging their strengths in feature extraction and pattern recognition, CNNs effectively analyze patterns in network traffic, such as abnormal packet flows or unusual access behaviors, allowing them to detect complex and subtle anomalies that signify APTs. By using their powerful feature extraction and pattern recognition capabilities, CNNs detect these intricate patterns and enhance the detection of sophisticated cyber threats.

Recurrent Neural Networks (RNNs) are another crucial type of deep learning model, that are adept at capturing temporal patterns in network traffic, such as time-series data, thereby enhancing the detection of APTs (Hewamalage et al. 2021). RNNs are specialized for sequential data, making them suitable for analyzing time-series data such as network traffic logs (Das et al. 2023). RNNs have internal memory states that enable them to retain information about previous inputs while processing current ones, effectively capturing temporal dependencies. Long Short-Term Memory (LSTM) networks, a type of RNN, are designed to handle long-term dependencies by using gates to control the flow of information, preventing the vanishing gradient problem commonly encountered in standard RNNs (Smagulova and James 2020). RNNs and LSTMs can model the sequential nature of network activities, such as user login sessions or data transfer patterns over time (Al-Selwi et al. 2024). This helps in identifying unusual sequences of events that could indicate an ongoing APT attack. For example, RNN-based IDSs can detect APTs by identifying actions that deviate from typical behavior, such as accessing sensitive files at odd hours, which could indicate a compromised account (Keshk et al. 2023).

Autoencoders (AEs) are also extensively utilized for anomaly detection by learning to represent normal behavior and subsequently identifying deviations from this learned behavior, thus enabling the detection of potential threats (Schneider et al. 2022). Autoencoders are effective for anomaly detection in network traffic, as they can identify unusual patterns that deviate from the learned normal behavior (Hdaib et al. 2024). This makes them useful for detecting subtle and rare event characteristic of APTs (Salim et al. 2023). For example, an autoencoder trained on normal network traffic can flag unusual login times or data access rates as anomalies, helping security analysts identify and respond to potential threats.

5.3.1 Transformer-based approach

Transformer architectures, which were originally developed for natural language processing, have been utilized for APT detection. This model can focus on important parts of the input sequence, thus enhancing the detection of relevant events and activities within large volumes of data. In recent advancements, (Zhang et al. 2023) proposed an intrusion detection method that leverages the strengths of both the Transformer and LSTM models. This combination results in robust intrusion detection with high accuracy and efficient processing capabilities. However, the complexity of integrating both models may pose challenges for real-time processing. (Ullah et al. 2023) developed IDS-INT, a system utilizing transformer-based transfer learning specifically designed for imbalanced network traffic. This method significantly improves the detection rates for minority classes, thereby addressing a common issue in network security. Nonetheless, the requirement for large volumes of labeled data presents a notable limitation, particularly in scenarios with imbalanced datasets. (Y.Liu and Wu 2023) focused on enhancing the standard transformer model to increase the intrusion detection performance. Their improvements resulted in significant accuracy gains; however, the computational intensity of the enhanced model may hinder real-time detection applications.(Y. Wang and Li 2023) proposed an anomaly-detection method for time-series data based on transformer reconstruction. This approach offers accurate and timely anomaly detection. However, managing high-dimensional time-series data remains challenging, which may impact overall performance.

(Ullah et al. 2022) introduced an explainable system using transformer-based transfer learning combined with multi-model visual representation. This approach enhances interpretability and accuracy; conversely, reliance on visual representation can result in increased computational overhead. (Z. Zhang and Wang 2022) proposed an efficient intrusion detection model that integrates CNNs with transformer models. This hybrid approach achieved high accuracy with reduced computational costs. Nevertheless, the complexity introduced by combining the CNN and Transformer models can complicate training and maintenance processes. (Huang et al., n.d.)applied a Transformer with TSGL for Named Entity Recognition (NER) in the cyber threat intelligence domain. This method improved the accuracy of the information extraction. The limitation is that this application may be restricted to the cyber-threat intelligence domain and may not be generalizable to other areas.

5.3.2 Training and validation

The data were split into training and validation sets. The models were trained on the training set, and their performance was validated on the validation set. Techniques such as cross-validation were used to ensure robust evaluation. These studies demonstrate the potential of transformer-based models for enhancing the effectiveness and interpretability of IDSs. Each method has unique strengths, while also presenting specific challenges that need to be addressed for broader application in real-world scenarios.

5.4 Black box model

Understanding the decision-making processes of AI is important for organizations. Monitoring AI models and ensuring accountability is essential rather than blindly trusting them. XAI aids in machine learning (ML), deep learning, and neural network algorithms. Many ML models are often viewed as black boxes (Chennam et al., 2023), as neural networks are extremely difficult to interpret. Additionally, biases in AI models and performance drift due to different production data pose significant risks. Therefore, organizations need to continuously monitor and manage the model to promote AI explainability, build trust; and reduce compliance, legal, security, and reputational risk.

5.5 XAI models

XAI models are crucial for interpreting the predictions of deep learning models used in APT detection. Techniques such as LIME, SHAP, and LRP provide insights into the model's decision-making process, enhancing transparency and trustworthiness. The details of this process are presented below.

5.5.1 Local interpretable model-agnostic explanations (LIME)

LIME creates local approximations of the model by perturbing the input data and fitting a simple, interpretable model to these perturbed instances, helping explain individual predictions by highlighting the specific features that influence the decision. LIME is widely used for interpreting local predictions (Hasan et al. 2023) by learning local linear approximations (Ibrahim et al. 2023). This algorithm provides local explanations of individual predictions (Volkov and Averkin 2023).

5.5.2 Shapley additive explanations (SHAP)

SHAP uses Shapley values from cooperative game theory to indicate the contribution of each feature to the model's predictions, providing a consistent measure of feature importance (Hasan et al. 2023). This technique determines which features are most influential in detecting APT activities and offers global explanations, thus helping cybersecurity experts understand the model's decision-making process. For example, SHAP can highlight unusual login times or data transfer volumes that might indicate an APT attack (Lundberg and Lee 2017). By assigning contribution values to each feature, their importance in decision making can be demonstrated (Band et al. 2023). This method, which relies on Shapley's principles from cooperative game theory, has been effective in various areas, including IDSs.

Specifically, SHAP proposes the following three methods:

- kernelSHAP: A model-agnostic version of the algorithm that operates only with the inputs and outputs of the function (Remman et al. 2021).
- treeSHAP: Computes the Shapley values on tree-based classification models (Wallsberger et al. 2022).
- deepSHAP: Optimizes the computation for neural network architectures (Meister et al. 2021).

By utilizing these methods, SHAP provides a comprehensive and interpretable framework for understanding and explaining model predictions.

5.5.3 Layer-wise relevance propagation (LRP)

The LRP computes relevance scores for individual input features by backpropagating the relevance of the class output node in a layer-wise fashion down to the input layer (Bach et al. 2015). This propagation adheres to a strict conservation property, ensuring that the relevance received by any neuron is redistributed equally. In CNNs, the LRP carries information about the relevance of the output class from higher layers back to the input pixel space layer by layer (Gu et al. 2019). By assigning relevance scores to input features, the LRP aims to interpret their contributions to model decisions. This technique requires access to the neural network's structure because it performs a backward pass through the network to compute relevance scores(Montavon et al. 2019). This method enhances the interpretability of neural networks by providing insight into the importance of individual features in the decision-making process of the model.

By integrating deep learning models with XAI techniques, we aimed to improve the detection accuracy and interpretability of APT detection systems. This approach ensured that cybersecurity experts could respond effectively to threats and understand the rationale behind specific decisions made by the models.

6 Explainable AI in cybersecurity

XAI is a type of artificial intelligence (AI) that focuses on the study and development of techniques to enable machines to behave in smart ways that humans can understand (Gunning, 2019). The goal is to create models that are easier to understand and more accurate. This allows people to trust and control their next-generation AI partners (Capuano et al. 2022a). The use of XAI techniques has gained significant attention in cybersecurity, especially for APT detection (Pawlicki et al. 2024).

XAI aims to provide transparency and explainability (Saeed and Omlin 2023). This shows how APT detection models make decisions so that cybersecurity experts can understand and trust the model's prediction (Haque et al. 2023). Moreover, XAI systems should respond to feedback from cybersecurity experts or decision-makers and adjust their actions accordingly. By incorporating XAI methods in APT detection systems, organizations can improve their defense strategies, enhance operational efficiency, and increase transparency and accountability. However, current XAI-related work in cybersecurity still lacks sufficient comparisons of different XAI algorithms in terms of metrics such as model specificity, scope, and methodology (Yang et al. 2023). Improving the coverage of detection system studies that leverage modern interpretability techniques will reinforce the technical depth of APT detection.

Figure 7 illustrates the timeline of the significant XAI review papers published from 2018 to 2024. This timeline provides a historical perspective on the development and evolution of XAI techniques, highlighting key contributions and advancements in the field. The exploration of XAI has garnered significant interest across various domains, particularly in enhancing the interpretability and transparency of complex machine learning models. This review delves into the contributions of researchers in advancing the field of XAI.

(Reis et al. 2019) developed a framework for explainable machine learning aimed at detecting fake news, focusing on both interpretability and accuracy. Their approach highlighted the necessity of transparent AI models to mitigate misinformation. (Xu et al. 2019) provided a brief survey of the history, research areas, approaches, and challenges of XAI, offering a foundational understanding of the field's evolution.

Insider threats pose significant cybersecurity challenges that common security solutions do not adequately address. (Homoliak et al. 2020) proposed a structural taxonomy and novel categorization to organize and clarify insider threat incidents and defense solutions. They utilized a grounded theory method for rigorous literature review, categorizing incidents, datasets, analysis, simulations, and defense solutions. Their taxonomy builds on existing frameworks and the 5W1H information gathering method. This survey aimed to enhance insider threat research by providing a structured taxonomy, an overview of publicly avail-

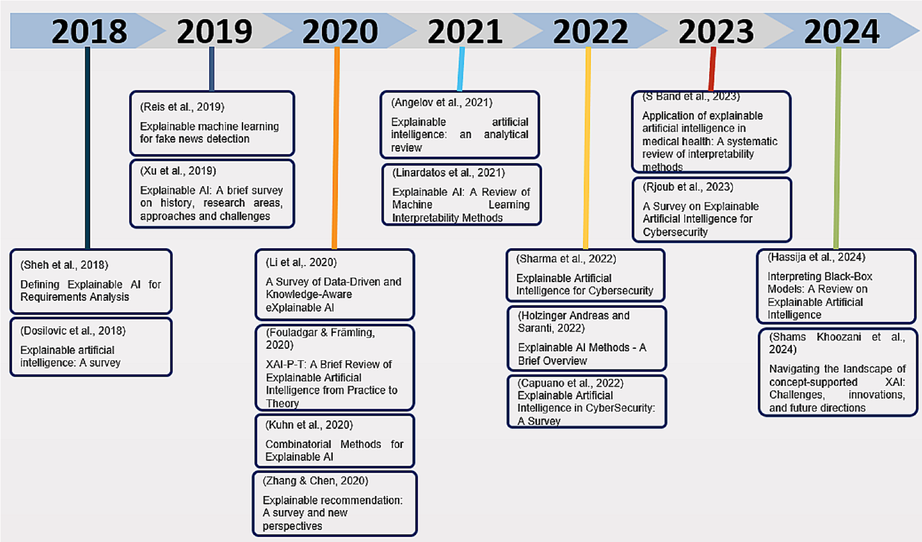


Fig. 7 Timeline of XAI review papers published from 2018 - 2024

able datasets, references to existing case studies and frameworks, and a discussion on trends and future research directions.

(Speith 2022) explored deep learning-based XAI concepts, proposed a taxonomy method, and created a timeline of key XAI studies. They demonstrated the need for structured approaches to navigate the rapidly growing field of explainable artificial intelligence (XAI). The recent surge in publications related to XAI has created a daunting challenge for those seeking to start or stay current with the latest developments. (Giudici and Raffinetti 2021)introduced the Shapley-Lorenz method for XAI, which focuses on fairness and transparency in AI decisions. Their work emphasized ethical considerations in AI model explanations. (Fouladgar and Främling 2020) reviewed both the practical and theoretical aspects of XAI and proposed a framework for integrating XAI into various applications. They bridge the gap between theory and practice, highlighting the versatility of XAI methods. (Kuhn et al. 2020) discussed combinatorial methods for enhancing the interpretability of AI models and provided insights into advanced techniques for model interpretation. (Angelov et al. 2021) presented an analytical review of XAI methods, critically evaluating their strengths and weaknesses. They offered a nuanced perspective on the effectiveness of different XAI techniques in various contexts. (Linardatos et al. 2021) reviewed machine learning interpretability methods, categorized them and discussed their applications across different domains. Their work emphasized the importance of methodological categorization in understanding XAI techniques.

(Sharma et al. 2022) explored the application of XAI techniques in cybersecurity, and proposed methods to enhance transparency and trustworthiness. They address the critical need for interpretable AI models in securing digital infrastructure. (Zhang et al. 2022) surveyed current literature on XAI techniques in cybersecurity, emphasizing the need for transparent and accountable models. They propose a clear roadmap for future XAI research in this domain. provided an overview of various XAI methods, summarizing their applications and effectiveness. This concise overview serves as a valuable reference for researchers and practitioners. (Capuano et al. 2022b)surveyed XAI techniques in cybersecurity, and proposed approaches to enhance interpretability and effectiveness. They highlighted the evolving landscape of XAI applications for securing digital environments.

(Rjoub et al. 2023) reviewed XAI techniques in cybersecurity, and proposed approaches for enhancing the interpretability of AI models. Their work focused on making AI systems more transparent and understandable. (Band et al. 2023) conducted a systematic review of interpretability methods in medical health applications, and proposed a framework for XAI in healthcare. They demonstrated the critical role of interpretability in clinical decision making. (Hassija et al. 2024b) provided a meticulous review and comprehensive analysis of state-of-the-art XAI models to address their complexity and lack of interpretability.. They offer insight into overcoming the challenges associated with complex AI systems. (Shams Khoozani et al. 2024) discussed the challenges, innovations, and future directions of concept-supported XAI, navigating the landscape of XAI research and application.

These contributions collectively advance the field of XAI by addressing key challenges, proposing innovative methods, and highlighting practical applications of XAI across various domains. This review highlights the importance of continued research and development on XAI to ensure the deployment of trustworthy and interpretable AI systems.

6.1 Explanation methods

Before delving deeper into the potential use of XAI for APT detection, we first briefly review XAI techniques. We investigated the most prominent, state-of-the-art XAI techniques. We analyzed the explanations provided by LIME, SHAP (Galli et al. 2024) LRP (Bach et al. 2015), and the attention mechanism. XAI techniques represent a diverse range of approaches. LIME focuses on local model explanations through perturbations of input data, whereas SHAP uses a game-theoretical approach to allocate contributions to each feature. The LRP propagates relevance through neural network layers, and attention mechanisms highlight the input features attended to by the model.

LIME and SHAP offer model-agnostic explainability, making them applicable to a wide range of models, including neural networks. Decision trees and rule-based models are inherently interpretable. (Sarker et al. 2024) proposed a decision tree-based model for intrusion detection, which allows for easy interpretation of detection rules.

The LRP specifically addresses layer-wise relevance in neural networks, and attention mechanisms are commonly used in neural networks for sequence data. LIME and SHAP have gained widespread acceptance in the research community due to their versatility and effectiveness in describing complex models. The LRP is renowned for its application in neural networks, and attention mechanisms have become standard in natural language processing and computer vision tasks.

6.2 Integration of XAI techniques in deep learning models

This paper thoroughly explores how XAI techniques, such as feature attribution and saliency maps, enhance deep learning models for APT detection (Lo et al. 2023). Specifically, it demonstrates how these approaches identify critical factors in the model's decision-making process. By incorporating SHAP, LIME, and LRP into the deep learning framework, this paper offers a comprehensive approach to improve interpretability. This integration clarifies influential factors and helps identify potential threats, thereby increasing the efficacy and transparency of cybersecurity measures.

This paper uses feature attribution methods to determine the importance of each factor in the model's predictions and employs saliency maps to visualize critical regions in network traffic data (Gevaert et al. 2024). By utilizing these XAI approaches, the interpretability of deep learning models can be enhanced. This transparency allows cybersecurity experts to understand and trust the model's predictions, making it easier to validate and fine-tune the detection systems (Sharma et al. 2022). Furthermore, integrating XAI helps pinpoint the factors most indicative of APT activity, improving detection accuracy and enabling quicker response times by focusing on the most relevant data aspects.

The paper includes practical examples and case studies demonstrating the application of XAI in real-world scenarios, illustrating how its use leads to more effective and transparent APT detection systems.

6.3 Comparative analysis of XAI- based APT detection

In recent years, the integration of XAI techniques with deep learning models has gained significant attention in the field of APT detection. This subsection provides a comprehensive overview of recent advancements from 2020 to 2024, highlighting various approaches and their effectiveness in enhancing cybersecurity measures.

(Singh et al. 2024) introduced SFC-NIDS, a sustainable and explainable network intrusion detection approach that analyzes VM traffic at the hypervisor level. It uses a gradient descent-based flow filtering mechanism, auto-encoders to reconstruct traffic features, and a 1D-CNN to detect malicious flows. When validated with hypervisor traffic artifacts and the KDD99 dataset, it achieves 98.9% and 99.97% accuracy, respectively. The strengths of these methods include high accuracy, adaptability, and explainability, but they are difficult to implement and specific to hypervisor environments, requiring further validation with various datasets.

(Khan et al. 2024) presented a novel security model for biomedical data collection and transmission, addressing privacy, security, and reputation concerns in medical networks. They introduced a threat-vector database based on the dynamic behaviors of smart healthcare systems and designed an improved SRU network to mitigate fading gradient issues and enhance the learning process by reducing computational costs. This approach is parallelizable and computationally efficient, dynamically adjusting the number of participating clients to reduce the communication overhead. Additionally, the model enhances the understanding of security experts by visualizing the decision process and explaining feature relevance. Compared with existing methods, this security model thoroughly analyzes and detects severe security threats with high accuracy, reduces overhead, lowers computational costs, and enhances the privacy of biomedical data. However, the potential complexity in implementing the novel security model across diverse healthcare networks and the need for further validation and testing in real-world healthcare environments to ensure robustness and effectiveness are noted as weaknesses.

Building on the idea of integrating explainability into AI-based cybersecurity solutions. (Galli et al. 2024) proposed a framework that integrates XAI methodologies within AI-based malware detection processes to address the lack of interpretability in traditional machine learning (ML) and deep learning (DL) approaches. The framework incorporates four XAI methods, such as SHAP, LIME, LRP, and Attention mechanisms. The key strengths of the proposed system include the integration of multiple XAI methods to improve model interpretability, thorough evaluation across diverse datasets, insightful comparisons of the LSTM and GRU models, and enhanced real-world applicability due to improved explainability. However, the study also faces challenges such as increased computational complexity, the need for further validation across different types of malware and cybersecurity contexts, and potential trade-offs between model explainability and performance.

Similarly, (Hasan et al. 2023) developed a highly effective model for identifying APTs using boosting-based machine learning methods, with XGBoost achieving an impressive weighted F1 score of 0.97. They enhance model interpretability by integrating SHAP, providing actionable insights for stakeholders. The key strengths of their work include exceptional predictive accuracy and enhanced transparency through XAI. However, potential challenges include computational complexity and the need for broader validation across various cyber threats. This approach underscores the promise of boosting-based XAI models in cybersecurity.

Expanding on the use of XAI in cybersecurity, (Zolanvari et al. 2023) employed TRUST XAI using multimodal Gaussian distributions and the TRUST Explainer technique. They evaluated their model on multiple datasets, including WUSTL-IIoT, NSL-KDD, and UNSW, and achieved 98% accuracy. A limitation of this study is that they did not address the computational resources necessary to implement their models.

(Khan et al. 2022) proposed a novel explainable deep learning framework for cyber threat discovery in Industrial IoT (IIoT) networks, addressing the critical need for data integrity and accuracy. They used an autoencoder-based detection system with convolutional and recurrent networks, employing a two-step sliding window technique to enhance feature extraction from raw time series data. Fully connected networks classify and explain attack events using temporal and spatial features. Empirical results demonstrated robust performance, with the framework outperforming contemporary methods. However, its complexity might limit scalability, and further exploration is needed for generalizability, real-time adaptation, and explainability depth.

(Khan et al. 2024) focused on bidirectional simple recurrent units (SRUs) and developed the XSRU-IoMT method, which was tested on the ToN_IoT dataset. Their model achieved 98% accuracy, but was validated on only a single dataset, limiting its generalizability. (Zhou et al. 2022) used an M&M Decision Tree model with prime implicant explanation techniques on various DDoS attack datasets, and achieved perfect recall and F1 scores. This method, which is based on an artificial immune system, may not be suitable for all intrusion-detection scenarios.

Another approach by (Patil et al. 2022) implemented an ensemble method using voting classifiers and LIME for interpretability on the CICIDS2017 dataset, achieving 96.25% accuracy. Reliance on a black box model may limit its transparency. (Barnard et al. 2022) combined extreme gradient boosting (XGBoost) with autoencoders (AEs) and utilized SHAP for interpretability, achieving 93% accuracy on the NSL-KDD dataset. However, this framework was assessed using only this dataset.

(Houda et al. 2022) applied deep neural networks (DNNs) with RuleFit, LIME, and SHAP on the NSL-KDD and UNSW NB-15 datasets, achieving 88% accuracy. However, their framework evaluation on only two datasets may not be generalizable to other datasets.

Le et al. (2022) used ensemble trees (DT and RF classifiers) and SHAP on NF-BoT-IoT v2 and NF-ToN-IoT-v2 datasets, and achieved 100% accuracy. However, the method's long training time might render it unsuitable for real-time IDS systems.

(Kuppa and Le-Khac 2021) explored autoencoders with counterfactual explanations on the Leaked Password Dataset, achieving 96.7% accuracy. Their study did not cover the broader aspects of machine learning or AI beyond cybersecurity. (Liu et al. 2021) proposed FAIXID using Boolean Rule Column Generation (BRCG) with data cleaning methods on real-world datasets, which achieved 87% accuracy. In this study, they did not provide detailed information on the size and diversity of the datasets used for the evaluation. Wali et al. (2021) utilized a primary Random Forest Classifier (RFC) with SHAP, achieving 98.5–100% accuracy on the Hop Skip Jump Attack and CICIDS datasets. The proposed framework requires significant computational resources.

(Antwarg et al. 2021) employed autoencoders with kernel SHAP on the KDD Cup 1999 and Credit Card datasets, focusing on explaining anomalies with a mean MSE of 0.0006. However, this method may not be applicable to APTs. Wang et al. (2020) used one-vs-all and multiclass classifiers with SHAP on the NSL-KDD dataset, achieving accuracies between 80.3% and 80.6%. The proposed framework was assessed using an outdated NSL-KDD dataset, which limited its relevance. Table 4 summarizes the XAI-based methods employed in cybersecurity.

This review of XAI-based APT detection methods highlights the diversity and advancements in the field while also identifying specific shortcomings that need to be addressed. These insights pave the way for future research to develop more robust, interpretable, and effective APT detection systems.

6.4 Case studies

This section presents various case studies that demonstrate the application of XAI techniques for combating APTs. These scenarios provide a comprehensive overview of how XAI can enhance the detection and mitigation of APTs.

Figure 8 compares different attack scenarios with the corresponding XAI techniques applied to improve detection and interpretation. The figure shows how techniques such as SHAP, LIME, LRP, and attention mechanisms enhance the interpretability of models in various attack scenarios such as phishing, insider threats, DoS attacks, and malware.

To address various attack scenarios, recent studies have applied XAI techniques to improve the detection and interpretation of cyber threats. (Adebowale et al. 2023) proposed integrating CNN and LSTM models with LIME to enhance the email classification model for phishing attacks. This hybrid XAI approach effectively addresses the challenges posed by large datasets and significantly improves classifier prediction performance.

(Homoliak et al. 2020) utilized SHAP to explain the global behavior of a detection model. Their study emphasized the diverse nature of insider threats and demonstrated how SHAP can identify common patterns and unique contributing factors across different clusters, such as fraud, IP theft, and sabotage. For DoS attacks, (Hariharan et al. 2021) applied Permutation Importance, SHAP, LIME, and Contextual Importance and Utility algorithms to provide a unified measure of feature importance and contribution to model predictions. Their case study revealed key insights into the impact of various features on IDS prediction performance, highlighting the most influential features for detecting DoS attacks.

Table 4 Summary of XAI-based methods used in cybersecurity

Author (s)	Application	Method	Dataset	Accuracy%	Precision%	Recall%	F1-Score%
(Singh et al. 2024)	Network Intrusion Detection	Gradient descent-based flow filtering, autoencoders, 1D-CNN	Virtual Network Traffic Artifacts, KDD99	98.9 and 99.97	-	-	99.03% and 99.98%
(Khan et al. 2024)	Biomedical Data Security	Threat-vector database, improved SRU network	Various healthcare datasets	-	-	-	-
(Galli et al. 2024)	Malware Detection	XAI methods (SHAP, LIME, LRP, Attention)	Diverse datasets	99.43	95.69	91.81	93.79
(Hasan et al. 2023)	APT Threat Detection	Boosting-based ML, XGBoost with SHAP	SCVIC-APT-2021	97	97	97	97
(Zolanvari et al. 2023)	Cybersecurity	TRUST XAI, multimodal Gaussian distributions	WUSTL-IIoT, NSL-KDD, UNSW	97.77–99.98	-	-	-
Moustafa, et al., 2022	Cyber Threat Discovery in IIoT Networks	Autoencoder, convolutional, recurrent networks	ToN_IoT	98	-	-	-
(Khan et al. 2022)	Cybersecurity in IoMT	Bidirectional SRU, XSRU-IoMT	ToN_IoT	99.38	99.39	98.99	99.37
(Zhou et al. 2022)	DDoS Attack Detection	M&M Decision Tree, prime implicant explanation	Various DDoS attack datasets	-	-	100	100
(Patil et al. 2022)	Intrusion Detection	Ensemble method, voting classifiers, LIME	CICIDS2017	96.25	89	89	89
(Barnard et al. 2022)	Network Intrusion Detection	XGBoost, autoencoders, SHAP	NSL-KDD	93	91	97	-
(Houda et al. 2022)	Network Intrusion Detection	DNN, RuleFit, LIME, SHAP	NSL-KDD, UNSW NB-15	88	89	87	88
Le et al., 2022	IoT Botnet Detection	Ensemble trees (DT, RF), SHAP	NF-BoT-IoT v2, NF-ToN-IoT-v2	100	-	-	-

Table 4 (continued)

Author (s)	Application	Method	Dataset	Accuracy%	Precision%	Recall%	F1-Score%
(Kuppa and Le-Khac 2021)	Password Leak Detection	Autoencoders, counterfactual explanations	Leaked Password Dataset	96.7	-	-	-
(Liu et al. 2021)	Intrusion Detection	FAIXID, BRCG, data cleaning methods	Real-world datasets	87	-	-	-
(Antwarg et al. 2021)	Anomaly Detection	Autoencoders, kernel SHAP	KDD Cup 1999, Credit Card datasets	-	-	-	-
(Wang et al. 2020)	Intrusion Detection	One-vs-all, multiclass classifiers, SHAP	NSL-KDD	80.3–80.6	-	-	-

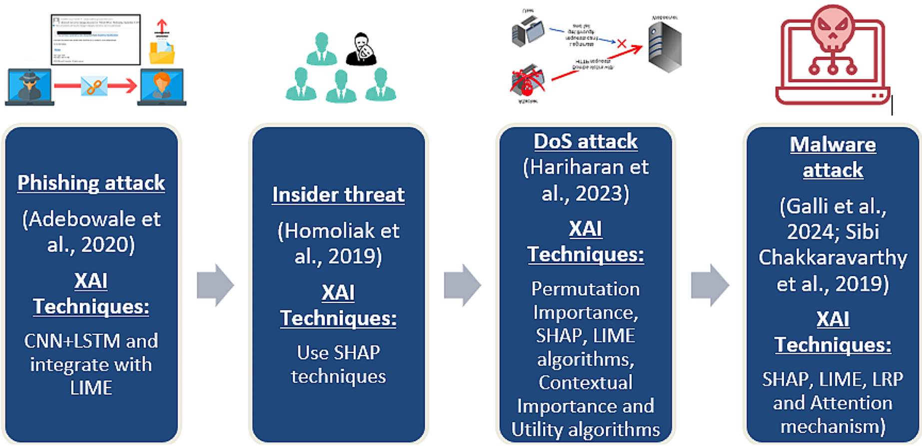


Fig. 8 Attack scenarios vs. XAI techniques

A downloader, a type of Trojan horse, downloads and installs malicious software without user consent. The explanation provided by the LRP highlights the Image 4 call, which attempts to copy files from one part of the file system to another, as discussed by (Galli et al. 2024). Finally, for malware attacks, (Galli et al. 2024) integrated the SHAP, LIME, LRP, and Attention mechanisms with robust deep learning models (e.g., LSTM and GRU) trained on labeled datasets. Their findings demonstrated that these XAI techniques significantly enhance the interpretability of malware detection models, ensuring that they remain effective and understandable in real-world applications.

Beyond cybersecurity, XAI has significant applications in other domains, such as the following:

- Healthcare providers have observed an increase in phishing attempts and suspicious activity in their networks. Given the sensitivity of patient data, they sought to improve their security measures by using an advanced detection system that combines deep learning and XAI techniques (Kalutharage et al. 2024).The data used include email traffic data collected from the providers' email servers, encompassing metadata and content analysis, as well as network logs from firewalls and intrusion detection systems (IDSs) that capture network activity and possible intrusions. CNNs can analyze email traffic and detect phishing attempts (McGinley and Monroy 2021), whereas RNNs can analyze sequential network logs for signs of APT activity. XAI techniques, such as SHAP, explain the phishing detection of CNN models, while LIME provides interpretability for network activity anomalies. The explanations provided by SHAP and LIME helped security teams identify specific email campaigns targeting employees and unusual data streams that indicated an APT in progress. This facilitated immediate corrective actions, including email filtering and network segmentation (Jia et al. 2021).

- A major financial institution experienced unusual network activity, suggesting a potential APT attack. The institution's security team decided to use an integrated approach that combines deep learning models and XAI techniques to identify and mitigate the threat. The data used included network traffic data collected from the institution's internal network, encompassing packet headers, payloads, and flow data over a six-month period. Additionally, system and application logs from critical servers and endpoints were analyzed, providing insights into user activities and system changes. SHAP values indicated that unusual login times and connections to known malicious IP addresses were significant factors in these detections. The LIME explanations for specific alerts revealed that certain endpoints repeatedly communicated with suspicious external servers, suggesting a potential APT foothold. These insights allowed the security team to prioritize these alerts and initiate a targeted investigation. The integrated approach not only detected the APT early but also provided clear explanations that helped the security team understand the attack vector and take swift action. As a result, the financial institution was able to isolate the affected endpoints, mitigate the threat, and prevent data exfiltration.

These use cases demonstrate how XAI techniques can be applied to various aspects of APT detection, from email classification and insider threat detection to network anomaly detection and malware identification. By examining these case studies, cybersecurity experts can better understand the practical applications of XAI and see how they can enhance their ability to investigate and mitigate APT attacks. These examples also highlight the potential benefits of XAI in real-world APT detection scenarios.

6.5 Role of XAI in APT detection

To effectively integrate XAI with deep learning for APT detection, several enhancements are necessary to address the unique challenges posed by sophisticated threats. (1) Handling sparse data is critical because traditional methods struggle with sparse datasets. Integrating XAI with anomaly-detection techniques, such as autoencoders, can help identify and explain anomalies in sparse data. (2) The high-dimensional nature of APT data requires dimensionality reduction techniques, such as Principal Component Analysis (PCA) or t-distributed Stochastic Neighbor Embedding (t-SNE), to simplify data complexity while retaining essential features, thereby improving interpretability. Moreover, APTs are characterized by evolving tactics, techniques, and procedures (TTPs). To adapt to these changes, XAI techniques must be dynamic and capable of real-time updating through continuous learning frameworks. This allows models to refine their understanding of new attack patterns and provide explanations that highlight these adaptations. (3) In terms of high-dimensional space, advanced feature importance techniques, such as Integrated Gradients, should be used to attribute importance across different layers of neural networks, offering a deeper understanding of model decisions. Finally, the implementation of a real-time feedback loop, in which cybersecurity experts can interact with the XAI system to refine explanations is crucial. This can be achieved using reinforcement learning to adapt models based on analyst feedback, ensuring that explanations remain accurate and relevant (Kute et al. 2021). To address sparse data, high-dimensional space, evolving tactics, and the need for real-time adaptation, this review aims to provide a more robust framework for explaining AI models used in APT detection, thereby advancing state-of-the-art models in cybersecurity.

In the discussion section, we explore the implications of these findings and how they contribute to advancing APT detection. Specifically, we discuss how the enhanced interpretability provided by XAI techniques can lead to more effective threat mitigation and greater trust in AI-driven security solutions.

7 Discussion and recommendations

This section delves into the need for explainability, challenges, and future directions in APT detection using deep learning and XAI. Traditional anomaly detection techniques are effective for known threats but struggle with new, evolving threats and large-scale data. Recurrent Neural Networks (RNNs) show promise in detecting complex and sequential patterns in network traffic, but they face challenges in interpretability and computational demands. XAI offers potential solutions by making model decisions transparent and understandable, adopting trust and collaboration between human experts and AI systems. This integration aims to increase the effectiveness and robustness of APT detection systems.

7.1 Key considerations for XAI

7.1.1 The need for explainability

XAI is crucial in making deep learning models interpretable and transparent, especially in cybersecurity where trust and reasoning are essential. Various techniques have been developed to enhance model interpretability. Feature importance methods, such as permutation importance and SHAP, determine the significance of each feature in the decision-making process (Hariharan et al. 2021). Partial Dependence Plots (PDPs) visualize the marginal effect of a feature on the predicted outcome (Ding et al. 2021). Activation Maximization identifies inputs that maximize the activation of specific neurons, revealing what each neuron "wants to see" (Zeltner et al. 2021). Saliency Maps highlight parts of the input most relevant to the model's decision (Brunke et al. 2020). LIME explains the predictions of any classifier by learning a simple model locally around the prediction (Houda et al. 2022). Layer-wise Relevance Propagation (LRP) backpropagates the output through the network to assign a "relevance score" to each input feature (Seibold et al. 2021). These techniques collectively enhance the transparency and trustworthiness of AI models, facilitating their adoption in critical cybersecurity applications.

7.1.2 Need for trust

Trust is essential in cybersecurity, and XAI methods enhance this by providing clear, understandable explanations of AI model predictions. The SHAP and LIME techniques help experts determine which features influence decisions, fostering confidence in AI systems. Transparent models reduce perceived risk and ensure fair, accurate operations by identifying and correcting biases (Saeed and Omlin 2023). This increased trust leads to better decision-making and faster response times, which are crucial in mitigating APTs. By integrating XAI, cybersecurity experts can confidently use AI systems, resulting in more robust and effective threat detection and response strategies (Sourati et al. 2023).

7.1.3 Need for reasoning

XAI enables cybersecurity experts to understand and validate model reasoning. This leads to better human-AI collaboration. This makes the detection of APT threats more effective. When security teams understand how the model works, they can better mitigate the potential vulnerabilities to adversarial attacks. This can enhance the robustness of the APT detection system more effectively. By seeing how the model makes decisions, security teams can identify and fix errors more efficiently. This helps improve the APT detection system.

XAI enables cybersecurity experts to understand and validate AI model reasoning, enhancing human-AI collaboration and APT detection (Sarker et al. 2024). By revealing how models arrive at their conclusions, XAI helps identify and mitigate vulnerabilities to adversarial attacks, improving system robustness. Understanding model decisions allows security teams to correct errors efficiently, maintaining detection accuracy (Došilović et al. 2018). XAI also fosters a shared understanding between AI systems and human operators, promoting the trust and effective use of AI in cybersecurity, ultimately leading to better detection and response strategies. The next subsection addresses the current issues associated with black-box models.

7.2 Current issues surrounding black-box models

One problem with using deep learning models for APT detection is their black box behavior, which makes it difficult to interpret and explain their decisions. (Kute et al. 2021). For example, when dealing with APT incidents, cybersecurity experts must understand why a particular event is deemed malicious. The lack of transparency in black-box models can lead to several issues. Cybersecurity experts may not trust a model's predictions without understanding the reasoning, especially when false positives or false negatives can have severe consequences. (Galli et al. 2024). When a model makes an incorrect prediction, identifying the root cause and making the necessary adjustments becomes challenging without insight into the model's decision-making process (Samek et al., 2017). During a security incident, organizations may face difficulties in explaining and justifying actions based on the model's predictions, potentially leading to legal and regulatory issues.

Additionally, black-box models are vulnerable to adversarial examples (Jia et al. 2024), which are carefully crafted inputs that deceive a model and lead to incorrect predictions. Without a clear understanding of how the model works internally, detecting and mitigating such attacks is more difficult (Ali et al. 2023).Therefore, promoting AI explainability is imperative to ensure the reliability and robustness of AI systems in critical applications such as cybersecurity.

Black box models pose significant challenges because of their opaque nature, making it difficult to interpret and explain their decisions. This lack of transparency can lead to issues such as biases, performance drifting, and vulnerability to adversarial attacks. In cybersecurity, the inability to understand and justify model predictions can undermine trust, complicate incident response, and lead to legal and regulatory challenges. XAI is essential for enhancing model transparency, building trust, and mitigating various risks, ensuring the reliability and robustness of AI systems in critical applications.

7.3 Research challenges and recommendations

Figure 9 shows that APT detection is technically challenging for several reasons. To address the challenges identified in APT detection using deep learning and XAI, several practical solutions are recommended.

Highly imbalanced datasets can hinder the performance of AI models in detecting minority class events. We address this issue using data augmentation techniques, such as the Synthetic Minority Oversampling Technique (SMOTE), to generate synthetic samples for the minority class(Y. Liu and Wu 2023). Employing unsupervised or semi-supervised learning techniques specifically designed to handle imbalanced data can also be effective (Zhang et al. 2020), (Kute et al. 2021)

The rapidly evolving threat landscape requires the continuous adaptation of detection models. To address this problem, continual learning approaches allow models to adapt to new data without forgetting previously learned information. Automated pipelines for data collection, model retraining, and deployment ensure that the detection system remains updated with the latest threat information (Khoozani et al., 2024). APT tactics, techniques, and procedures (TTPs) are continuously changing, making it difficult for static models to remain effective. By integrating XAI with deep learning models, we enhance understanding and adapt to new attack patterns. This integration improves the robustness of the system against evolving threats, ensuring that it remains effective in detecting and mitigating APTs.

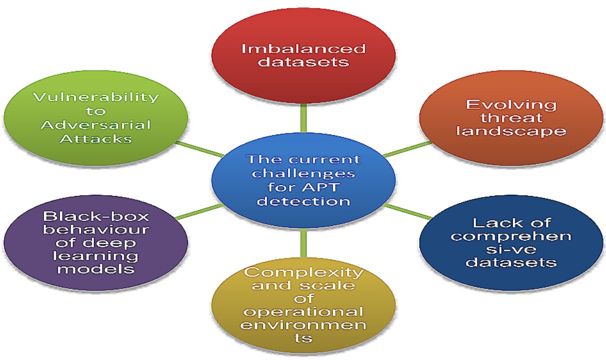


Fig. 9 Research challenges for APT detection

Comprehensive datasets for effective APT detection are lacking. This problem can be addressed through collaborative efforts among academia, industry, and government agencies to create large-scale, diverse, and labeled datasets. Promoting data-sharing initiatives encourages organizations to share anonymized threat data for research purposes, thereby enriching the pool of available datasets (Ferrag et al. 2020).

Integrating XAI techniques into APT detection systems is challenging because of the complexity and scale of operational environments. (Kuhn et al. 2020) developed modular XAI frameworks that can be easily integrated into existing cybersecurity infrastructures. In addition, providing training and resources for cybersecurity experts to use XAI tools effectively enhances their practical applicability.

Deep-learning models are often criticized for their difficulty in interpretation and lack of transparency (Liang et al. 2021). XAI techniques, such as SHAP and LIME, highlight the features that most influence model predictions and provide visual explanations that are easy to interpret. AI models must provide clear, understandable explanations for their decisions to build trust and enable cybersecurity experts to make informed decisions (Galli et al. 2024).

Black-box models can be tricked by adversarial examples, which can lead to incorrect predictions. Understanding the internal operation of the model is crucial for detecting and mitigating such attacks. The development of XAI systems that reveal the decision-making process of the model can help identify and counter adversarial inputs effectively (Kenny et al. 2021).

Our discussion highlights the strengths and potential of integrating XAI with deep learning for APT detection. The enhanced interpretability provided by XAI techniques allows cybersecurity experts to understand and trust model predictions, leading to more effective threat mitigation. In conclusion, we summarize our contributions and suggest directions for future research, emphasizing the need for continued innovation in this critical area of cybersecurity.

8 Conclusions and future research directions

Our research addresses the challenges in APT detection by integrating XAI with deep learning models. As illustrated in Fig. 10, future research directions for APT detection focus on several key areas. First, examining the potential of deep feature extraction methods is essential for enhancing detection capabilities. Second, integrating XAI techniques with deep learning models is crucial, as it offers transparent and effective explanations for APT detection. Additionally, optimizing automatically extracted features can enhance detection accuracy, reduce human bias, and improve resource efficiency. Furthermore, augmenting existing datasets by generating synthetic examples of APT activities and benign behaviors can enrich the data available for training. Moreover, it is important to evaluate detection models using comprehensive evaluation metrics and scenario-based security testing to ensure robust performance. Finally, maintaining privacy and security is essential when robust detection models that safeguard data privacy are constructed, thus ensuring the overall integrity of the system.

This review provides a comprehensive overview of state-of-the-art models for dvanced persistent threats (APTs) detection. Through extensive analysis, we found that deep learn-

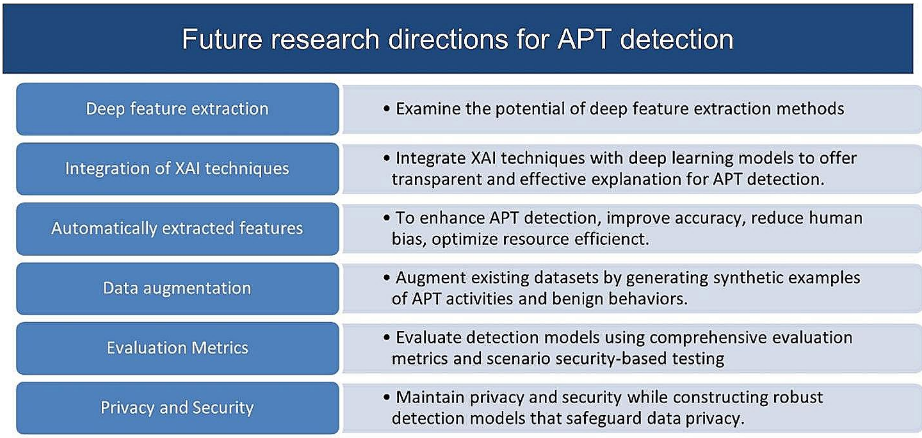


Fig. 10 Future research directions for APT research

ing-based techniques, especially those incorporating XAI methods are increasingly prevalent. These approaches address significant challenges in traditional IDSs, such as low detection accuracy, high false-positive rates, and difficulties in detecting unknown or early-stage attacks. By incorporating XAI techniques such as SHAP, cybersecurity experts can understand the contribution of each feature to the model's prediction, such as unusual login times or unexpected IP addresses, allowing for quicker verification of alerts and reduced false positives. Moreover, the transparency provided by XAI models enhances trust and accountability in detection systems. Users and stakeholders can audit and validate the decisions made by the models, ensuring that the detection process is transparent and reliable. Empirical evidence from recent studies further supports the effectiveness of integrating XAI with deep learning, demonstrating improvements in detection accuracy and a significant reduction in false positives. This not only validates our approach but also underscores its practical benefits in real-world applications.

Our review also highlights the strengths and weaknesses of various APT detection methods across different datasets, techniques, and experimental results. By identifying gaps and challenges in current methodologies, we propose integrating XAI with deep learning models to advance the state-of-the-art. This integration not only improves detection accuracy and scalability but also provides clear, interpretable insights into the model's decision-making process, ensuring effective threat response. The impact of our findings is substantial for cybersecurity. By enhancing the transparency and interpretability of APT detection systems, we can enable more effective and timely responses to sophisticated threats. This research contributes to the development of advanced security measures that adapt to evolving cyber attacker tactics.

In conclusion, our research paves the way for more robust, scalable, and interpretable APT detection systems that effectively combat sophisticated APT attacks. By addressing both technical and interpretability challenges, we advance the state-of-the-art in cybersecurity and enhance network resilience to APTs.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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